TFPT — Origin Theory

The seam as a horizon, the cyclic compiler hull,
and the self-reproducing parameter-free fixed point

An [E] structural core (`v53`–`v56`) with one honestly-typed [C] interpretation

Stefan Hamann　　Alessandro Rizzo

July 14, 2026 · v5.4 (rev 367)

What this document is, and is not

This note collects, in one derivation, why the two TFPT inputs leave *no free dimensionless compiler dial beyond the anchor structure and* π (the one dimensionful scale $v_{\rm geo}$ and the continuous transfer physics $F_{\rm transfer}$ remain explicitly typed, not derived), and how the entire integer skeleton, the seam constant $c_3 = \frac{1}{8\pi}$ and a built-in cyclic structure all flow from one boundary pair. **Two layers, kept strictly apart:**

- **Structural core** [E] (§1–§5): exact, machine-checked identities (`v53`–`v56`, plus `v6`/`v8`/`v3`). These stand on their own, independently of any cosmology.
- **One interpretation** [C] (§6): the *cyclic self-reproduction* reading — collapse → horizon=seam → next cycle. This is a physical hypothesis, *not* derived and *not* machine-checkable; it is offered because it gives the structural fixed point a physical selection mechanism. It is flagged [C] throughout.

Nothing in the [C] layer is sold as proven; nothing in the [E] layer depends on the [C] layer.

Marker key: ***[E]*** *exact / machine-proven ·* ***[C]*** *conditional (holds under named hypotheses) ·* ***[O]*** *open / axiom ·* ***[X]*** *falsifiable kill test. Per-claim fine type in* `verification/status_ledger.csv`.

Contents

1 The whole integer skeleton from one pair $(g_{\rm car}, N_{\rm fam}) = (5, 3)$ **4**

2 The "8" is triply forced: geometry = lattice = gravity 10

3 A cyclic element *inside* the hull: the order-30 Coxeter rotation 11

4 The unique attractor: a gapped transport selects ONE fixed point 12

5 One α^{-1} **for electromagnetism, the cosmic horizon and** Λ **18**

6 Interpretation [C]: the self-reproducing cosmic cycle 19

7 Falsifiers 20

8 The seam and the horizon: the precise statement 22

9 Consequences, and the whole story [C] 26

10 A closure program (next phase) [O] 27

11 Honest status 29

The TFPT document set — what is treated where

Plain language: TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\rm car} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the

Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and three companions**, best read in order (with **four** companions):

0. `introduction` — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). *(entry point)*

1. `tfpt_1_architecture_e8` — the two axioms, the derivation map, the EM fixed point α^{-1}, the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction, the QBL theorem chain.

2. `tfpt_2_standard_model` — the SM in one φ_0-ladder formula, flavor from parabolic transport (sheet diamond, dual anchor, Q_+ cohomology), the worked closures.

3. `tfpt_3_e8_audit_bootstrap` — the seven E_8 slices as an audit raster, the cascade bridge, the Möbius bootstrap.

4. `tfpt_4_frontier` — honest status of η_B, m_p/m_e, Koide, dark matter, full QG.

5. `tfpt_5_redteam` — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

Companions: R. `tfpt_research_contracts` — the open interfaces ($v_{\text{geo}}, G_{\text{net}}, F_{\text{transfer}}$) as numbered contracts; **H.** `tfpt_horizon_readouts` — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** `origin_theory` — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation); **S.** `tfpt_safeguards` — the verification discipline: every mechanism that defends a claim against coincidence and numerology (status calculus, no-free-pattern, the over-determination map, the firewall, frozen predictions + null model, the independent Wolfram and Lean paths, the red team).

You are reading companion O (Origin Theory).

TFPT status at a glance — read this card the same way in every document

Two inputs, one compiler. From the boundary constant $c_3 = \frac{1}{8\pi}$ (axiom P1) and the five-slot carrier $g_{\rm car} = 5$ (axiom P2) — jointly the elementary symmetric data of one anchor $a = (1, 1, 2)$ — the discrete compiler $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ($\alpha^{-1} = 137.0359992$) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

What is closed, and what genuinely remains. The discrete/algebraic layer is closed ([E]); the everything-open residual reduces to *three named interface problems*:

Interface	Question	Status
$v_{\rm geo}$	the one metrology unit ($= 1/\sqrt{G} = m/\mu$); No-Unit Thm: no compiler scale	primitive [O]
$G_{\rm net}$	simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle(1, 1)\rangle \cong (E_8)_1$	[C]
$F_{\rm transfer}$	one functor, four typed interfaces (Koide, η_B, axion, m_p/m_e)	[C]

Gravity is parameter-free. The classical metric-sector field equation is no longer only an $R+R^2$ readout: the fixed-volume entanglement equilibrium (Jacobson; Faulkner et al.), run with TFPT's atoms, gives the *full* covariant Einstein equation $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ with *both* coefficients fixed — $c_3^{-1} = 8\pi$ (no free dimensionless Newton dial; the thermodynamic origin $2\pi/\eta$ *coincides* with the geometric one $|\mathbb{Z}_2| 2\pi\chi$ via $|\mu_4| = |\mathbb{Z}_2|\chi(S^2) = 4$, so c_3 is triply over-determined — anchor, geometry, thermodynamics, `v358`) and Λ from α ($\rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$, `v60`); the Einstein tensor (not Ricci) is forced by Lovelock, so matter conservation is an output (`v359`). The residual here is only the equation-of-state status (an external candidate action — Bianconi's entropic action, arXiv:2408.14391 — is quantified in `v473` without changing the typing) and the one unit $v_{\rm geo}$; the global measure (C7 = `QG.AMB.01`) is now a [C] redundancy (`v369`), a certification object rather than missing dynamics, conditional on `SEAM.EQUIV.01` and Bisognano–Wichmann.

The seam-net structural theorem $G_{\rm net}$ (= `SEAM.EQUIV.01`: the raw seam *is* the holomorphic $(E_8)_1$ net) is now *closed modulo a cited theorem*. The explicit lattice model (`v367`/`v368`) and the S_3 closure stack pin the target at every computable level — central charge $c = 8$ (`v376`), the $(E_8)_1$ character with 248 currents and one primary (`v377`), genus-one torus GSD = 1 (`v378`) and reflection positivity (`v379`) — and it is Lean-formalised as `FORM.SEAM.MMST.01`: the collar's MMST hypotheses are kernel-proved, the MMST scaling-limit and Adamo–Moriwaki–Tanimoto OS-reconstruction theorems enter as named cited axioms, and the `#print axioms` check is clean. The *only* residual that stays [O] is the abstract continuum existence of the scaling limit (`v336`) — its 128-spinor extension leg since certified at net level by the peer-reviewed crossed-product package (h_s=16/16=1∈$\mathbb{Z}$, the Longo–Rehren locality criterion; Böckenhauer; KLM μ=4/2^2=1; AGT/AMT the independent second witness) with the realisation input reduced to invariant level and the parallel Lean route `seamResidualClosed′` (`v469`); so this is *closed modulo a cited theorem*, not solved. `QGEO.SYM.01` (the μ_4 deck acting geometrically) is its *corollary*: a conformal net's vacuum is rotation-invariant by axiom, so the deck follows with no extra premise (`v335`, Lean `FORM.QGEO.BW.01`). The nonperturbative quantum-gravity measure `QG.AMB.01` is no longer carried as an open item: it is a [C] *redundancy* (`v369`), a certification object rather than missing dynamics, conditional on `SEAM.EQUIV.01` and Bisognano–Wichmann (and TFPT is moreover rigorously gap-decoupled from the general Euclidean-QG conformal-factor problem, margin $1.648 > 0$, `v76`/`v330`). So the honest residual is the one unit $v_{\rm geo}$ plus the typed $F_{\rm transfer}$ interfaces, above the cited-theorem ceiling on the seam and with `QG.AMB.01` a [C] redundancy. Everything carries a claim ID resolving to a row of `verification/status_ledger.csv` (the single source of truth); **if the text and the ledger ever disagree, the ledger wins.** The historical labels ($U_{\rm wall}, G_{\rm metric}, F_{\rm frontier}$) are retained only for ledger continuity — the live residual is $v_{\rm geo} \oplus F_{\rm transfer}$, with the seam $G_{\rm net}$ closed modulo a cited theorem and `QG.AMB.01` a [C] redundancy.

Marker key: ***[E]*** *exact / machine-proven ·* ***[C]*** *conditional (holds under named hypotheses) ·* ***[O]*** *open / axiom ·* ***[X]*** *falsifiable kill test. Per-claim fine type in* `verification/status_ledger.csv`.

[E] exact / machine-proven [C] conditional [O] open / axiom [X] kill test

P1 $c_3 = \frac{1}{8\pi}$ **P2** $g_{\text{car}} = 5$ *(two inputs, [O] axioms)*

↓

Anchor $a = (1,1,2)$: $e_1, e_2, e_3 = 4, 5, 2$ *([E] exact)*

↓

Discrete compiler: $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$, $h = 30$, φ_0 *([E] exact)*

↓

SM skeleton: three families, hypercharges, flavor matrix R *([E])*

↓

Physical bridges: v_{geo}, G_{net}, F_{transfer} *([C] bridges)*

↓

Frontier readouts: Λ, inflation, η_B, dark matter, QG *([C]/[O])*

1 The whole integer skeleton from one pair $(g_{\text{car}}, N_{\text{fam}}) = (5, 3)$

TFPT runs on two operational inputs: the seam constant $c_3 = \frac{1}{8\pi}$ (P1) and the carrier rank $g_{\text{car}} = 5$ (P2). The boundary $\mu_4 = \{1, i, -1, -i\}$ gives the four-punctured line $\mathbb{P}^1 \setminus \mu_4$ with $H_1 = \mathbb{Z}^3$, hence the family count $N_{\text{fam}} = 3$ (A_3). We show the rest of the discrete data is generated by the single pair $(g_{\text{car}}, N_{\text{fam}}) = (5, 3)$.

One object, two canonical invariants. The pair $(g_{\text{car}}, N_{\text{fam}})$ is read off the *same* four-point divisor $D = \mu_4$ (degree 4) by the two canonical functors on $\mathbb{P}^1$: the cycle side rank $H_1(\mathbb{P}^1 \setminus \mu_4) = \deg D - 1 = 3 = N_{\text{fam}}$ and the function side (Riemann–Roch) $h^0(\mathbb{P}^1, \mathcal{O}(\mu_4)) = \deg D + 1 = 5 = g_{\text{car}}$, the latter matched by $\text{rank}(D_5) = 5$ (the carrier is the $\mathfrak{so}(10)$ half-spinor). Reading the carrier as this meromorphic seam-mode space (at most simple poles on μ_4) is geometrically natural [E] in its arithmetic, though the physical identification stays [C] and does not displace the over-determined $g_{\text{car}} = 5$ (Pascal/bootstrap/Lean) [`verification/v189_riemann_roch_carrier.py`].

And the function side generates D_5 itself, not just g_{car}. The even Clifford algebra of the 5-dimensional mode space closes the chain $\mu_4 \to g_{\text{car}} \to D_5$: $\Lambda^{\text{even}}(C_{\text{car}}) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 1 + 10 + 5 = 16 = 2^{g_{\text{car}}-1} = \dim S^+$, which is exactly the $\mathfrak{so}(10) = D_5$ half-spinor. So the carrier is not merely "rank 5 like D_5" — it *is* the Clifford spinor of the Riemann–Roch mode space [E], with the physical identification again [C] [`verification/v197_rr_carrier_clifford_d5.py`].

The $(5,3)$ generator and the Pythagorean mass volume [E] ([`verification/v53_compiler_core.py`])

By sum, difference and halving,

$$\text{rank}\, E_8 = g_{\text{car}} + N_{\text{fam}} = 8, \qquad |\mathbb{Z}_2| = g_{\text{car}} - N_{\text{fam}} = 2, \qquad |\mu_4| = \tfrac{g_{\text{car}} + N_{\text{fam}}}{2} = 4,$$

and $(N_{\text{fam}}, |\mu_4|, g_{\text{car}}) = (3, 4, 5)$ is *the* Pythagorean triple. This is load-bearing: the grand-mass-volume exponent $\Delta_Y = g_{\text{car}}^2 = 25$ ($\det M_{\text{SM}} \sim (\varphi_0)^{25}$) splits as a *difference of squares*,

$$\boxed{\Delta_Y = g_{\text{car}}^2 = \underbrace{N_{\text{fam}}^2}_{\text{down}=9} + \underbrace{(g_{\text{car}} - N_{\text{fam}})(g_{\text{car}} + N_{\text{fam}})}_{=|\mathbb{Z}_2| \cdot \text{rank}\, E_8 = \dim S^+ = 16} = 9 + 16 = 25}$$

(the K-row sums $(6, 9, 10)$: down carries N_{fam}^2, up+lepton carry $|\mathbb{Z}_2| \cdot \text{rank}\, E_8$). The *anchor* $a = (1, 1, 2)$ is the unique 3-multiset whose elementary symmetric functions are the compiler atoms $(|\mu_4|, g_{\text{car}}, |\mathbb{Z}_2|) = (4, 5, 2)$; equivalently $\chi_a(t) = (t-1)^2(t-2) = t^3 - |\mu_4| t^2 + g_{\text{car}} t - |\mathbb{Z}_2|$.

The two glue discriminant norms are likewise $(g_{\rm car}, N_{\rm fam})$ over the glue index: $q(D_5) = g_{\rm car}/|\mu_4| = \frac{5}{4}$, $q(A_3) = N_{\rm fam}/|\mu_4| = \frac{3}{4}$, so their sum is the E_8 root norm 2, their difference is the lepton transport value $\delta = |\mathbb{Z}_2|/|\mu_4| = \frac{1}{2}$, their product is $\dim \mathfrak{su}(4)/\dim S^+ = \frac{15}{16}$ and their ratio is $g_{\rm car}/N_{\rm fam} = \frac{5}{3}$ [`verification/v51_boundary_half_step.py`]. The single transcendental is π; both π-coefficients are 2π times a compiler integer.

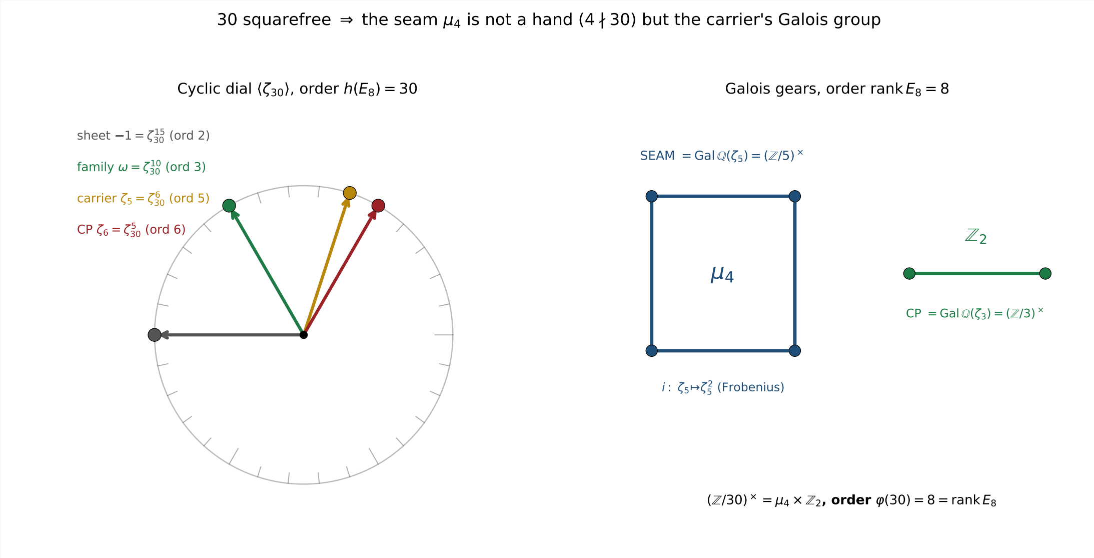
The icosahedral bedrock: *why* the atoms are $\{2,3,5\}$ [E] ([`verification/v219_icosahedral_mckay.py`])

The pair $(g_{\rm car}, N_{\rm fam}) = (5,3)$ and the sheet $|\mathbb{Z}_2| = 2$ are not three free small integers: $\{2,3,5\}$ is forced as soon as one fixes E_8. By the **McKay correspondence** the finite subgroups of $SU(2)$ are classified by the affine ADE diagrams, and E_8 is the exceptional *maximum* of that spherical tower ($2T\to\hat{E}_6$, $2O\to\hat{E}_7$, $2I\to\hat{E}_8$). Its branch data is the *icosahedron*: the binary icosahedral group $2I$ (order $120{=}|R^+(E_8)|$) has element orders $\{1,2,3,4,5,6,10\}$ — it literally *contains* the $2,3,5$ rotation-axis orders — and its nine irreducible representations have dimensions $\{1,2,2,3,3,4,4,5,6\}$, the Kac marks of $\hat{E}_8$, with

$$\sum_i d_i = 30 = h(E_8) = |\mathbb{Z}_2|\cdot N_{\rm fam}\cdot g_{\rm car} = 2\cdot3\cdot5, \qquad \sum_i d_i^2 = 120 = |R^+(E_8)|.$$

The McKay graph is *built from the group* (the natural 2-dim rep gives the affine E_8 adjacency, top eigenvalue 2, with marks = degrees), so the recurring $\{2,3,5\}$ signature has a single bedrock: E_8 *is* the icosahedron. This is a *backward certificate* of the already-closed E_8 [E]; the reading "it explains the $(2,3,5)$ choice" is [C]; the raw-seam origin (without first positing E_8) stays the [O] P2 interface. The same exceptional geometry has two metric readings via complex multiplication ([`verification/v222_cm_norm_duality.py`]): on the *square* seam ($j{=}1728$, Gaussian $\mathbb{Z}[i]$) the EM index is the Gaussian norm $N(g_{\rm car}{+}i|\mu_4|) = 5^2{+}4^2 = 41 = 10b_1$, while on the *hexagonal* partner ($j{=}0$, Eisenstein $\mathbb{Z}[\omega]$) the scalaron exponent is $N(3{+}2\omega) = 7$ — the carrier split $(3,2)$ generates $(5,6,7,13)$ by sum/product/Eisenstein/Gauss norm, the rings rigid. The same two rings give the atoms a *splitting trichotomy* ([`verification/v416_atom_trichotomy.py`]): in $\mathbb{Z}[i]$ (the square seam) $2{=}|\mathbb{Z}_2|$ ramifies, $3{=}N_{\rm fam}$ is inert and $5{=}g_{\rm car}$ *splits* as $(2{+}i)(2{-}i)$, while in $\mathbb{Z}[\omega]$ (the hexagonal partner) 3 ramifies and $2,5$ are inert — so the ramified prime of each ring *is* its own atom, and the three quadratic facets $\mathbb{Q}(i), \mathbb{Q}(\sqrt{-3}), \mathbb{Q}(\sqrt{5})$ of the order-30 field $K = \mathbb{Q}(i,\sqrt{-3},\sqrt{5})$ (v403) ramify over exactly one atom each ($2,3,5$; product $30 = h(E_8)$), two imaginary (CM/dynamic) plus one real (RM/static). The arithmetic is [E], the seam/flavor/golden reading [C]. The same three rings carry the carrier split $(3,2)$ as one determinant triple ([`verification/v418_cyclotomic_norm_triple.py`]): $\det(3I + 2\,\mathrm{Comp}(\Phi_n)) = (7,13,55)$ for $n = 3,4,5$ over $\mathbb{Z}[\omega], \mathbb{Z}[i], \mathbb{Z}[\zeta_5]$ — the scalaron, Δ_Q and the quark numerator $55 = g_{\rm car}\|\,\mathrm{Pl}(K)\|_1$; the carrier-5 clock missing from the 3-dim flavor space is the $4{\times}4$ Φ_5-companion (with the golden φ as its real-part data). The anchor $(5,4)$ fails the triple in the carrier ring (461, prime), and honestly $(3,2)$ is the *forced* split (v14), not the unique clean one (the scan also finds $(2,1){\to}(3,5,11)$, a secondary rung). The sharper discriminator is *cross-sector*: only $(3,2)$ lights up three different physics sectors — gravity (7), flavor (13) and masses (55) — whereas $(2,1){\to}(3,5,11)$ is all compiler-core $(N_{\rm fam}, g_{\rm car}, \|\,\mathrm{Pl}(K)\|_1)$; the two rungs read as input $(2,1)$ → output $(3,2)$. [E]/[C]. And the seam μ_4 clock is the order-4 character of the E_8 Coxeter cycle itself: $(\mathbb{Z}/30)^\times$ are the E_8 exponents, with the order-4 generator 7 acting as a literal μ_4 clock on the eight live phases ([`verification/v223_coxeter_totative_clock.py`]).

The seam μ_4 is the carrier's Galois group ([`verification/v419_seam_galois_carrier.py`]). The deeper reason the seam sits on the Galois side, not on the cyclic dial, is that $h(E_8) = 30$ is *squarefree*: $\mathbb{Z}/30$ has no order-4 element, so μ_4 cannot be a rotation hand. By CRT $(\mathbb{Z}/30)^\times = (\mathbb{Z}/2)^\times{\times}(\mathbb{Z}/3)^\times{\times}(\mathbb{Z}/5)^\times = \mu_4{\times}\mathbb{Z}_2$ (order $\varphi(30) = 8 = \mathrm{rank}\,E_8$), and the μ_4 factor *is* $(\mathbb{Z}/5)^\times = \mathrm{Gal}(\mathbb{Q}(\zeta_5))$ — the carrier prime 5. So the seam μ_4 is the carrier pentagon's automorphism group: the Frobenius $\zeta_5 \mapsto \zeta_5^2$ is the explicit order-4 operator G with $G\,C_5\,G^{-1} = C_5^2$, $G^4 = I$ on the carrier clock C_5 (v418). This is the positive content of `RES.COXETER.SYMMETRY.01` (v409): the $2^2 = 4$ is the order of $(\mathbb{Z}/5)^\times$, and the seam "carries no rate" because it is an *automorphism*, not a hand. So $\mathbb{Q}(\zeta_{30})$ carries both E_8 invariants — the cyclic hands ($h = 30$) and the Galois gears (rank $= 8 = \mu_4\times\mathbb{Z}_2$).



The Galois gearbox IS the $(E_8)_1$ glue ([`verification/v422_galois_net_bridge.py`]). This bridges the Galois side directly to $G_{\rm net}$: the seam μ_4 is not merely *a* cyclic $\mathbb{Z}/4$, it is the *same* one as the holomorphic-net glue. The carrier discriminants $\mathrm{disc}(A_3) = \mathrm{disc}(D_5) = \mathbb{Z}/4$ are cyclic (one Smith invariant factor 4), and D_5 is cyclic precisely because its rank $5 = g_{\rm car}$ is *odd* (D_n is $\mathbb{Z}/4$ for n odd, $\mathbb{Z}_2 \times \mathbb{Z}_2$ for n even); the carrier-Galois $\mathrm{Gal}(\mathbb{Q}(\zeta_5)) = (\mathbb{Z}/5)^\times$ is cyclic $\mathbb{Z}/4$ for the same reason (5 prime); and in $\mathrm{disc}(D_5 \oplus A_3) = \mathbb{Z}_4 \times \mathbb{Z}_4$ the simple current $(1,1)$ generates the cyclic $\mathbb{Z}/4$ whose Lagrangian quotient $16/4^2 = 1$ *is* $(E_8)_1$ (v89/v154). The cyclic-vs-Klein distinction is load-bearing: the order-2 halfway glue gives $D_8 = SO(16)$ ($\det 4$), not E_8 ($\det 1$), so the carrier-Galois (cyclic, 5 prime) matches E_8, never the D_8 Klein. So the order-30 *gearbox* $(\mathbb{Z}/30)^\times$ is the simple-current glue of the seam net — one $\mathbb{Z}/4$ from carrier arithmetic through the lattice to the seam ([**E**]; the canonical identification [**C**]). This is the *four-fold μ_4 identity* — deck/divisor = Galois = discriminant = simple-current, one cyclic $\mathbb{Z}/4$ read four ways — and, by the over-determination accounting (v428), it counts as *compression* (a unification that makes the carrier less axiomatic), not four independent witnesses that multiply. Finally, P2 itself reads as a Riemann–Roch index: a degree-4 seam divisor on $\mathbb{P}^1$ forces $h^0{=}5{=}g_{\rm car}$ and rank $H_1{=}3{=}N_{\rm fam}$, with even Clifford closure $\Lambda^{\rm even}(\mathbb{C}^5) = 16 = \dim S^+$ ([`verification/v228_rr_index_gate.py`], [**E**]/[**C**]/[**O**], conditional on the seam producing the degree-4 divisor = `QGEO.SYM.01`). And the same icosahedron gives the seam a *geometric* home ([`verification/v232_e8_kleinian_seam.py`]): on the du Val side of McKay, the minimal resolution of the Kleinian singularity $\mathbb{C}^2/2I$ has an exceptional locus of *eight* rational curves glued in the E_8 pattern (their negated intersection form is the E_8 Cartan, $\det 1$; each a (-2)-curve), and its link is the Poincaré homology 3-sphere $S^3/2I$ ($2I$ is perfect, so $H_1 = 0$, $\pi_1 = 2I$ of order $120 = |R^+(E_8)|$). The eight curves are rank $E_8 = g_{\rm car} + N_{\rm fam}$ — a *fourth* reading of the "8" as the eight seam components. So the raw seam ($\mathbb{P}^1$ with marks) is canonically one of the E_8 du Val exceptional curves: `QGEO.REALIZE.01` acquires an algebraic-geometric model [**C**], though it still presupposes E_8 (the bedrock stays [**O**]).

The capstone: one singularity generates the skeleton ([`verification/v236_brieskorn_capstone.py`]). All of the above is the Milnor fibration of a *single* singularity — the $(2,3,5)$ Brieskorn singularity $x^2+y^3+z^5$, whose three exponents *are* the atoms $(|\mathbb{Z}_2|, N_{\rm fam}, g_{\rm car})$. Its Milnor number $\mu = (2-1)(3-1)(5-1) = 1{\cdot}2{\cdot}4 = 8 = \mathrm{rank}\,E_8$ is a *fifth* independent origin of the "8" (after geometry, lattice, gravity and $\varphi(30)$); its Milnor monodromy has eigenvalues the eight primitive 30th roots $e^{2\pi i m/30}$ with m the E_8 exponents (characteristic polynomial Φ_{30}), i.e. it *is* the order-30 E_8 Coxeter element (§3); its Milnor lattice is E_8 and its link is the Poincaré sphere. Both clocks are facets of this one monodromy: the μ_3 family/triality (the CP phase, [`verification/v233_cp_triality_phase.py`]) is the order-3 power h^{10} in $\langle h\rangle = \mathbb{Z}/30$, while the μ_4 seam clock ([`verification/v223_coxeter_totative_clock.py`]) is the order-4 Galois automorphism of $\mathbb{Q}(\zeta_{30})$. So the recurring $\{2,3,5\}$ is not many numbers but *one* icosahedral singularity read several ways [**E**].

The atoms are the network's spectral angles — the golden ratio is the $g_{\rm car}{=}5$ signature ([`verification/v313_golden_atoms_spectral.py`]). The same $(2,3,5)$ has a spectral face. The affine-E_8 adjacency characteristic polynomial factors as $x(x^2-4)(x^2-1)(x^2-x-1)(x^2+x-1)$, so its nontrivial eigenvalues are exactly $2\cos(\pi/k)$ for $k \in \{|\mathbb{Z}_2|, N_{\text{fam}}, g_{\text{car}}\} = \{2,3,5\}$: $2\cos(\pi/2){=}0$, $2\cos(\pi/3){=}1$, $2\cos(\pi/5){=}\phi$ (the golden ratio, with the second harmonic $2\cos(2\pi/5){=}1/\phi$). [E] So the golden ratio is *specifically* the $g_{\text{car}}{=}5$ (5-fold) signature, and $g_{\text{car}}{=}5$ gains a new spectral witness ($\pi/5$) distinct from Pascal and Riemann–Roch. The selection is the icosahedral one: $1/|\mathbb{Z}_2|{+}1/N_{\text{fam}}{+}1/g_{\text{car}} = 31/30 > 1$ is the spherical-triangle condition forcing the *finite* $2I \Rightarrow E_8$, and its numerator $31 = 2^{g_{\text{car}}}{-}1 = 248/8 = 1{+}h(E_8)$ is exactly the gap-decoupling margin numerator (v63/v76) over the Coxeter $30 = h(E_8)$ — a unification of already-anchored quantities, not new evidence. Honestly, ϕ is a *structural* lattice/network quantity, absent from the dimensionless SM readouts — with the sole [C] exception that $\cos(\theta_i{=}3\pi/5) = -1/(2\phi)$ on the axion spine branch, where θ_i is the regular g_{car}-gon (pentagon) interior angle ([`verification/v429_axion_pentagon_phi.py`]); and as the E_8-quasicrystal (Elser–Sloane cut-and-project) ratio it is a concrete candidate substrate for the rewrite question [C].

A minimal substrate that works ([`verification/v324_hypergraph_fiber.py`]). The rewrite question can be made concrete with a *fibred* $(2,3,5)$ network. The pure $(2,3,5)$ network gives only the Coxeter skeleton and the golden 5-fold angle; the recovery rate $(2/3)^6$ is provably *not* in its adjacency spectrum (v312). But the *smallest* substrate that carries everything is a simple *product*: the carrier network $T_{\text{net}} = (A{+}2I)/4$ (whose attractor is the Kac marks = the E_8 skeleton) fibred by a 3-node family cusp $T_{\text{cusp}} = \operatorname{diag}((1{-}w)^{2N_{\text{fam}}})$, $w \in \{0, \frac{1}{3}, \frac{2}{3}\}$ (whose spectrum $\{1, (2/3)^6, (1/3)^6\}$ supplies the recovery gap). [E] The fibred operator $T_{\text{net}} \otimes T_{\text{cusp}}$ carries *both* at once: its top eigenvalue 1 has eigenvector marks $\otimes\,(w{=}0)$ (the skeleton $\times$ the democratic family law), and $(2/3)^6$ appears as a genuine eigenvalue with eigenvector marks$\otimes(w{=}\frac{1}{3})$ (network attractor $\times$ cusp subleading). [E] This is the cyclotomic split made dynamical: substrate = carrier $\times$ family = the $30 = g_{\text{car}}(2N_{\text{fam}}) = 5\times 6$ of v315, the recovery gap living entirely in the family factor, and the minimal fibre the 3-node cusp ($= N_{\text{fam}}$). Honestly (v312) the cusp weight is still *injected*, not derived from the graph — so this is a consistent realisation of the predicted product structure, not a derivation of $2/3$ from a pure rewrite [C].

The cusp weight from a minimal rewrite ([`verification/v327_hypergraph_rewrite.py`]). One honest step further: the injected $2/3$ is the fixed point of a *minimal local rule*. A family channel with $N_{\text{fam}}{=}3$ slots, one absorbing attractor ($w{=}0$) and $|\mathbb{Z}_2|{=}2$ surviving (the $\mathbb{Z}_2$ pair), has per-step survival operator $M = \left(\begin{smallmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{3} & \frac{1}{3} \\ 0 & \frac{1}{3} & \frac{1}{3} \end{smallmatrix}\right)$ with spectrum $\{1, \frac{2}{3}, 0\}$ — so the recovery survival $\frac{2}{3} = (N_{\text{fam}}{-}1)/N_{\text{fam}} = |\mathbb{Z}_2|/N_{\text{fam}}$ *emerges from the rule arity*, and over the order-6=$2N_{\text{fam}}$ hand the rate is $(2/3)^6$. [E] Moreover $2/3$ is provably *not* a root of the affine-E_8 characteristic polynomial ($p(2/3) = -3520/19683 \neq 0$), so the recovery rate cannot be an adjacency eigenvalue (the v312 negative, now a proof). [O] The residual shrinks: the rewrite derives the *ratio* $2/3$ from the arity $\{|\mathbb{Z}_2|, N_{\text{fam}}\} = \{2,3\}$, leaving only the anchor atoms as input — so the substrate is the $(2,3,5)$ adjacency (carrier) $\times$ this minimal branching rule (family), both built from $\{2,3,5\}$ [C].

One coupled local rule ([`verification/v395_hypergraph_coupled.py`]). The three hypergraph modules ([`verification/v299_hypergraph_0d_autonomous.py`], [`verification/v327_hypergraph_rewrite.py`], [`verification/v324_hypergraph_fiber.py`]) merge into a *single* rewrite on a $9{\times}3$ labelled grid: one micro-step = network lazy diffusion on each cusp column + the [`verification/v327_hypergraph_rewrite.py`] rule M on each node's family vector ($T_{\text{micro}}{=}T_{\text{net}} \otimes M$, purely local on 27 cells). [E] The joint attractor is marks$\otimes(w{=}0)$; one clock hand ($2N_{\text{fam}}$ family micro-steps) carries recovery $(2/3)^6$; [`verification/v324_hypergraph_fiber.py`]'s $T_{\text{net}} \otimes T_{\text{cusp}}$ is the same substrate in the Galois/cusp readout basis ([`verification/v317_galois_family.py`]); [`verification/v299_hypergraph_0d_autonomous.py`]'s growth $E_6{\to}E_7{\to}E_8{\to}\hat{E}_8$ is unchanged with the fiber attached. [O] The mechanism is unified; the irreducible inputs remain the 3-arm seed (P_2) and the analytic seed φ_0 (puncture term still [`verification/v312_hypergraph_substrate.py`]).

The seed at leading order ([`verification/v396_phi0_icosahedral.py`]). The tree-level retained seed $\varphi_0^{\text{tree}} = 1/(6\pi)$ is not an independent hypergraph injection: on the icosahedral $(2,3,5)$ skeleton it equals $F/(4h\pi) = F/(|\operatorname{Aut}|\pi) = (F/(g_{\text{car}}N_{\text{fam}}))c_3$ with $F = 20$ triangular hyperedges, $h = 30$, $|\operatorname{Aut}| = 120$, and $F/h = |\mathbb{Z}_2|/N_{\text{fam}} = 2/3$ (the same ratio as [verification/v327_hypergraph_rewrite.py]). **[E]** Gauss–Bonnet consistent ($c_3 = 1/(|\mathbb{Z}_2|4\pi)$, [verification/v216_marks_gauss_bonnet.py]/[verification/v342_em_ward_heatkernel.py]). **[E]** The puncture $48c_3^4$ is still analytic, not a rational hypergraph fraction ([verification/v312_hypergraph_substrate.py]); it belongs to the *boundary* engine, not the graph — named `HYP.PHI0.PUNCTURE.01` ([verification/v408_phi0_puncture_heatkernel.py]): the order-$|\mu_4|$=4 local boundary heat-kernel (Seeley–DeWitt) contact term $\Omega_{\mathrm{adm}} c_3^4$ of the μ_4 puncture divisor, summed over Ω_{adm}=$N_{\mathrm{fam}} \dim S^+$=48 admissible states ($c_3$ the same Gauss–Bonnet datum, so c_3^4 a fourth-order boundary power; cf. v391). **[C]** Two engines, two terms: the icosahedral hypergraph delivers the tree term, the boundary heat-kernel delivers the puncture; its from-first-principles heat-kernel proof is the tracked **[O]** residual.

Static vs. dynamic rates: a number-field split ([verification/v314_rate_translation.py]). The discrete kernel and the dynamic recovery sit in *different number fields*, which is the precise reason their rates do not freely interconvert. The adjacency spectrum splits as a rational part $\{0, \pm1, \pm2\} \subset \mathbb{Q}$ and a golden part $\{\pm\phi, \pm1/\phi\} \subset \mathbb{Q}(\sqrt{5})$, while the dynamic transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ is *entirely rational*. **[E]** So the family/sheet rates *do* translate — the rational angles $2\cos(\pi/2)$=0 ($|\mathbb{Z}_2|$) and $2\cos(\pi/3)$=1 (N_{fam}) are the same atoms that build the recovery rate $(2/3)^6 = (|\mathbb{Z}_2|/N_{\mathrm{fam}})^{2N_{\mathrm{fam}}} \in \mathbb{Q}$ — but the carrier 5-fold (golden, $\mathbb{Q}(\sqrt{5})$) has *no* rational dynamic image (a field obstruction: 2/3 is neither an adjacency nor a lazy-update eigenvalue, and the heat-kernel time $\Delta/(2-\phi) \approx 6.37$ is not clean). **[E]** The carrier is therefore *static-only* — g_{car}=5 builds the lattice and spectrum, while $\{|\mathbb{Z}_2|, N_{\mathrm{fam}}\}$=$\{2, 3\}$ drive the relaxation. The only object holding both is the order-30 Coxeter element, $30 = g_{\mathrm{car}}\,(2N_{\mathrm{fam}}) = 5 \cdot 6$ (the golden in the order-5 factor, the recovery exponent 6=$2N_{\mathrm{fam}}$ in the order-6 factor); correspondingly the ϕ-quasicrystal $E_8 = H_4 + \phi H_4$ carries the carrier but not the (rational, 3-fold) cusp-weight family dynamics **[C]**.

Couple or factorize? A cyclotomic answer ([verification/v315_coxeter_coupling.py]). The order-30 Coxeter element does *not* dynamically entangle the two sectors — as a cyclic group it *factorizes*, $\mathbb{Z}/30 = \mathbb{Z}/5 \times \mathbb{Z}/6$ (a direct product, decoupled, in line with the field obstruction above). It couples them instead at the *field* level: $\mathbb{Q}(\zeta_{30})$ is the compositum of the carrier field $\mathbb{Q}(\zeta_5)$ (which contains the golden $\sqrt{5} = \phi + 1/\phi$) and the family field $\mathbb{Q}(\zeta_3)$, with degree $[\mathbb{Q}(\zeta_{30}){:}\mathbb{Q}] = 8 =$ rank E_8. **[E]** Its Galois group is exactly $(\mathbb{Z}/5)^\times \times (\mathbb{Z}/3)^\times = \mathbb{Z}/4 \times \mathbb{Z}/2 = \mu_4 \times \mathbb{Z}_2$ (order 8, maximal element order 4, *not* cyclic) — so the μ_4 seam clock *is* the Galois group of the carrier's golden field, and the $\mathbb{Z}_2$ sheet that of the family field. The carrier ($\sqrt{5}$, 5-fold) and the family (ζ_3, 3-fold) are thus conjugate sub-data of *one* cyclotomic field: the static↔dynamic bridge is *arithmetic* (Galois), not a dynamical generator **[C]**.

Does the Galois action reach the physics? The CP phases ([verification/v316_galois_readout.py]). It does — through the *family* factor. The hexagonal CP unit $\rho = \zeta_6 = \zeta_{30}^5$ is exactly the order-$6 = 2N_{\mathrm{fam}}$ factor c^5 above (the carrier factor being $\zeta_5 = \zeta_{30}^6$), so the leading CP phases are its cyclotomic arguments: $\delta_{\mathrm{CKM}}^{\mathrm{lead}} = \pi/3 = \arg\zeta_6$ and $\delta_{\mathrm{PMNS}} = 4\pi/3 = \arg\zeta_6^4$ (the power $4 = |\mu_4|$), with $\zeta_6^4 = -\zeta_6$ the $\mathbb{Z}_2$ sheet flip and $\mu_6 = \mu_3 \times \mu_2$ (family × sheet) **[C]**. **[E]** Crucially the family Galois factor itself is physical: $\mathrm{Gal}(\mathbb{Q}(\zeta_6)/\mathbb{Q}) = \mathbb{Z}_2$, $\zeta_6 \mapsto \zeta_6^5 = \bar\zeta_6$, *is* CP conjugation $\delta \mapsto -\delta$. The carrier factor ζ_5 (the golden $\sqrt{5}$) carries no phase — μ_4 stays static — and the *magnitudes* ($\sin^2\theta_{12} = \frac{1}{3} - \varphi_0/2$, with φ_0 transcendental) are not cyclotomic at all. So the $\mu_4 \times \mathbb{Z}_2$ coupling touches the readouts exactly in the phase sector the field split predicts: the CP phases are family-factor cyclotomic data, the magnitudes the analytic seed **[E]**.

Are the three generations themselves a Galois orbit? ([verification/v317_galois_family.py]). The same arithmetic governs the family count, not only the phase. The three cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\}$ (with $\mathrm{Spec}(Q_+) = 3\alpha + 1 = \{1, 2, 3\}$, the A_3 exponents) are, under $w \mapsto e^{2\pi i w}$, the cube roots of unity $\{1, \zeta_3, \zeta_3^2\}$, and the cyclic family symmetry $w \mapsto w + \frac{1}{3}$ is a *transitive* μ_3 orbit — the three generations *are* the cube-root orbit. **[E]** The family Galois $\mathbb{Z}_2$ ($\zeta_3 \mapsto \zeta_3^2 = \bar\zeta_3$, the same CP conjugation as above) refines this to 1+2: it *fixes* the rational weight $w = 0$ and *swaps* $w = \frac{1}{3} \leftrightarrow \frac{2}{3}$. And the fixed (rational) generation is precisely the recovery *attractor* — $w = 0$ carries the transfer eigenvalue 1 (the stationary "law", §4), while the Galois-conjugate pair carries the decaying modes $(2/3)^6, (1/3)^6$ (the hierarchy). **[E]** So the family structure is the same arithmetic as the CP phase:

a μ_3 orbit, Galois-refined 1+2, the fixed point the selected law; only the mass *magnitudes* remain the analytic seed [C].

Capstone — what is genuinely free ([verification/v318_arithmetic_capstone.py]). Collecting the arc: the entire SM *structural* sector — the three generations, the two CP phases, their orbit/hierarchy ordering — is the cyclotomic field $\mathbb{Q}(\zeta_{30})$ with Galois group $\mu_4 \times \mathbb{Z}_2$ (degree $[\mathbb{Q}(\zeta_{30}){:}\mathbb{Q}] = 8 = \operatorname{rank} E_8$, $30 = |\mathbb{Z}_2| N_{\text{fam}} g_{\text{car}} = h(E_8)$), and this skeleton is *forced* by the anchor atoms $\{2,3,5\} = e(a)$ — no freedom. [E] The *magnitude* layer is no freer: the seed $\varphi_0 = (|\mu_4|/N_{\text{fam}})c_3 + \Omega_{\text{adm}} c_3^4 = \frac{4}{3}c_3 + 48c_3^4$ is a pure function of π (the coefficients anchor-derived), so $\sin^2\theta_{12} = \frac{1}{3} - \varphi_0/2$ and its kin are $\{a,\pi\}$-determined transcendentals — not cyclotomic, not free. [E] Hence the complete input is $\{a, \pi, v_{\text{geo}}\}$: *zero* dimensionless free parameters (everything is $\{a,\pi\}$), π the unique transcendental primitive, and v_{geo} the single dimensionful unit (No-Unit Theorem). [O] This rigidity holds *given* the axioms; it does not by itself realise the structure on the raw seam — that remains the open bedrock (`QGEO.SYM.01`/`SEAM.EQUIV.01`).

The translation clock: discrete ↔ dynamic is one clock, 5 × 6 [E]/[C] ([verification/v319_translation_clock.py])

There is a vivid way to read the whole arc above. The bridge between the *static* (lattice/spectrum) data and the *dynamic* (recovery/relaxation) data is not a map but a *clock* — the order-30 Coxeter element — and, because $\gcd(g_{\text{car}}, 2N_{\text{fam}}) = \gcd(5,6) = 1$, it factorises into *two coprime hands*:

$$\boxed{\mathbb{Z}/30 \;=\; \underbrace{\mathbb{Z}/5}_{\text{static carrier } g_{\text{car}}} \;\times\; \underbrace{\mathbb{Z}/6}_{\text{dynamic family } 2N_{\text{fam}}}}, \qquad 30 = h(E_8) = g_{\text{car}}\,(2N_{\text{fam}}).$$

The dynamic hand runs $0,1,2,3,4,5$ ($\mathbb{Z}/6 = 2N_{\text{fam}}$): the recovery exponent is *exactly* $6 = 2N_{\text{fam}}$, the rate $(2/3)^6 = (|\mathbb{Z}_2|/N_{\text{fam}})^{2N_{\text{fam}}}$, the order-6 generator c^5 ($\zeta_6 = \zeta_{30}^5$, the hexagonal CP unit). This is the hand that *translates* into a dynamic rate (the rational/family sector, v314). **The static hand runs** $1,2,3,4,5$ ($\mathbb{Z}/5 = g_{\text{car}}$): the golden field $\mathbb{Q}(\sqrt{5})$ ($\varphi = 2\cos(\pi/5)$), field-obstructed — it builds the lattice and spectrum but has *no* rational dynamic image, so it is static-only ($\zeta_5 = \zeta_{30}^6$ carries no phase). **And the "0 versus 1" is meaningful:** the resummed clock $\text{rate}(n) = -p_2 \ln(1 - n/N_{\text{fam}})$ (v124) has $\text{rate}(0) = 0$ — position 0 is the *conserved law* (the attractor, eigenvalue 1, §4) — while $n = 1, 2, \ldots$ are the live decaying modes; correspondingly the E_8 live phases (the Coxeter exponents = totatives of 30) *start at* 1, the 0 phase being the fixed law, not a totative. So $0,1,2,3,4,5$ is the law-inclusive reading and $1,2,3,4,5$ the live-only reading of the *same* hand. One full turn $= \operatorname{lcm}(5,6) = 30$ synchronises both hands: rank $E_8 = 8 = \varphi(30)$ live phases, pairing into $|\mu_4| = 4$ invariant planes (§3). The arithmetic is [E]; the reading "the bridge *is* one clock" is [C].

Falsifiable consequence: locked CP phases ([verification/v320_galois_cp_relation.py]). The arithmetic is not only descriptive — it makes a testable cross-prediction. Both leading CP phases are powers of the *one* hexagonal unit $\rho = \zeta_6$ of the family factor: $\delta_{\text{CKM}}^{\text{lead}} = \arg\rho = \pi/3$ ($60°$) and $\delta_{\text{PMNS}} = \arg\rho^4 = 4\pi/3$ ($240°$), and since $\rho^4 = -\rho$ they are *locked*,

$$\delta_{\text{PMNS}} \;=\; \delta_{\text{CKM}}^{\text{lead}} + \pi.$$

[C] So the previously *assigned* value $\delta_{\text{PMNS}} = 240°$ is upgraded to a Galois-forced relation to the *leading* ($\pi/3$) component of the quark phase: the quark and lepton leading CP phases are not independent. (The lock is to the structural $\pi/3$, *not* the full measured $\gamma = \delta_{\text{CKM}}^{\text{lead}} + 3\lambda_C^2 \approx 68.7°$ that carries the quark transport correction — so $\delta_{\text{PMNS}} = 240°$, not $248.7°$.) [X] This is a clean kill test — a δ_{PMNS} robustly incompatible with $\delta_{\text{CKM}}^{\text{lead}} + \pi = 240°$ (beyond the sub-leading budget, $> 3\sigma$ at DUNE/Hyper-K/JUNO) would falsify the whole Galois-CP organisation; $240°$ is currently within the broad global fit.

Sharpened: the node, the budget, the tension ([verification/v322_cp_lock_sharpened.py]). The lock is more than a relation. [E] The leading phase can only sit on the six hexagonal nodes $\arg\rho^k = k \cdot 60°$, and *which* one is selected is fixed by the deck order: $\delta_{\text{PMNS}}^{\text{lead}} = |\mu_4|\,\delta_{\text{CKM}}^{\text{lead}} = 4(\pi/3)$ (node 4). [C] The unclosed sub-leading correction is bounded by the quark analogue $3\lambda^2 \approx 8.7°$ (the full quark phase being $\delta_{\text{CKM}}^{\text{full}} = \pi/3 + 3\lambda^2 \approx 68.7°$), so the prediction is the narrow band

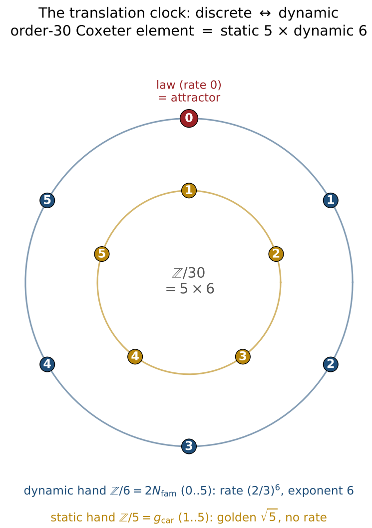


Figure 1: The translation clock ([`verification/v319_translation_clock.py`]): the discrete↔dynamic bridge is the order-30 Coxeter element read as two coprime hands. The *dynamic* hand ($\mathbb{Z}/6 = 2N_{\mathrm{fam}}$, outer, ticks 0..5) carries the recovery rate $(2/3)^6$ (exponent 6), with position 0 the conserved law (the attractor) and 1..5 the live phases; the *static* hand ($\mathbb{Z}/5 = g_{\mathrm{car}}$, inner, ticks 1..5) carries the golden $\sqrt{5}$ and no rate. One full turn is $\mathrm{lcm}(5,6) = 30 = h(E_8)$. [E] (arithmetic) / [C] (the one-clock reading).

$240^\circ \pm \sim 9^\circ$. [C] Against the current global fit (NuFIT 6.0, normal ordering, $\delta_{CP} = 212^{+26}_{-41}{}^\circ$) the leading value sits at $+1.08\sigma$ and the band edge at $+0.74\sigma$ — comfortably alive today. [X] The discrimination is clean: the nearest *wrong* node (180° or 300°) is 60° away, far beyond the budget, so a DUNE/Hyper-K measurement at its projected ~ 5–15° precision lands decisively on or off node 4 (an anarchic phase would hit the band only $\sim 5\%$ of the time, and the specific node 1-in-6).

2 The "8" is triply forced: geometry = lattice = gravity

The integer 8 in $c_3 = \frac{1}{8\pi}$ is not a free choice. Three independent routes give it, and the seam=horizon reading requires their agreement — which holds, overdetermined. **Firewall (status of the "8").** The geometry and lattice routes are exact [E]. The gravity route is an *exact arithmetic alignment* with the Hawking/Einstein 8π normalisation, hence also [E] *as an alignment*. What is *not* [E] is the physical statement "the seam *is* a horizon": that principle stays [C] until it is derived inside the theory (the Seam–Horizon Theorem, §8). The exact alignments below are [E]; the physical identification is [C].

One boundary transport for flavor *and* horizon [E] ([`verification/v54_seam_horizon_keystones.py`])

The boundary transport spectrum is $\{1, (2/3)^6, (1/3)^6\}$ (cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\}$ at the four μ_4-punctures). Its sub-leading eigenvalue $\lambda_2 = (2/3)^6$ appears in *both* sectors:

$$\text{SM flavor gap} \quad \Delta_{\mathrm{gap}} = -\log(2/3)^6 = 6\log\tfrac{3}{2} = 2.4328$$

$$\Longleftrightarrow \qquad \text{horizon Page recovery} \quad I_n \sim \lambda_2^n = (2/3)^{6n}.$$

One boundary operator governs the Standard-Model flavor hierarchy and the horizon information recovery — exactly what "the SM is the boundary data of the seam=horizon" would require.

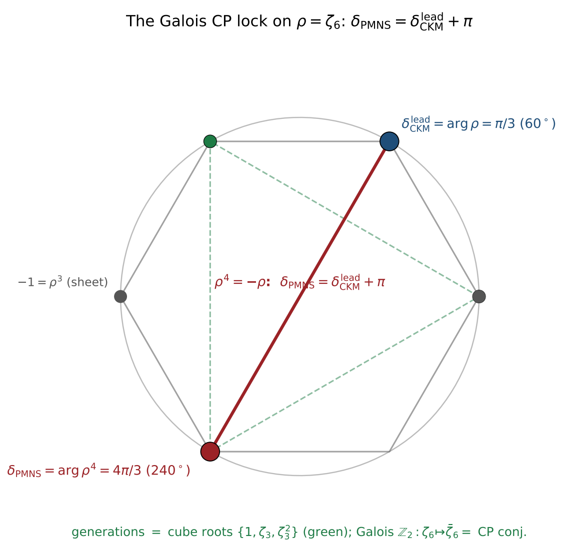


Figure 2: The Galois CP lock ([`verification/v320_galois_cp_relation.py`]): both leading CP phases are powers of the one hexagonal unit $\rho = \zeta_6$ — $\delta^{\rm lead}_{\rm CKM} = \arg\rho = \pi/3$ (blue) and $\delta_{\rm PMNS} = \arg\rho^4 = 4\pi/3$ (red) — locked by $\rho^4 = -\rho$ to $\delta_{\rm PMNS} = \delta^{\rm lead}_{\rm CKM} + \pi$. The three generations are the even vertices (cube roots $\{1, \zeta_3, \zeta_3^2\}$, green, §4); the family Galois $\mathbb{Z}_2$ ($\zeta_6 \mapsto \bar\zeta_6$) is CP conjugation. [E] (relation) / [C] (phase identification) / [X] (kill test).

The maximal black hole is the anchor [E] ([`verification/v101_horizon_anchor.py`])

The seam=horizon reading now has a *classical-gravity* fingerprint with zero free parameters. Put a black hole into the de Sitter bulk (Schwarzschild–de Sitter): the maximal case (Nariai, two horizons merging) has, in units $1/\sqrt{\Lambda}$, the horizon cubic

$$\boxed{t^3 - 3t + 2 = (t-1)^2(t+2): \ \text{roots } (1, 1, -2)}$$
$$= \text{ the \textit{traceless projection} of the anchor } a = (1, 1, 2),$$

with coefficients $(-N_{\rm fam}, |\mathbb{Z}_2|)$ and $M_N = 1/(N_{\rm fam}\sqrt{\Lambda})$. Each Nariai horizon carries $S_{dS}/N_{\rm fam}$; the total is $\frac{2}{3}S_{dS}$ — **the Koide branch value $2/3 = |\mathbb{Z}_2|/N_{\rm fam}$ is the exact entropy bound of the maximal black hole**. The interpolation is $S_{\rm tot}/S_{dS} = (x^2+1)/\Phi_3(x)$ (the $N_{\rm fam}$ cyclotomic), the three-root entropy total $\pi\sum r_i^2 = |\mathbb{Z}_2|S_{dS}$ is conserved for every M, the mass line is itself a split double cover (branch points $\pm 1/N_{\rm fam}$, separation $2/3$, deck = horizon swap — the gravitational twin of the flavor sheet $\mathbb{Z}_2$), and the evaporation flow runs *away* from the anchor point toward the democratic endpoint — the same repeller/attractor orientation as the flavor relaxation (§4). Six independent landings on already-load-bearing atoms; the GR identities are textbook, the bulk reading stays [C]. Details and figures: Appendix H.

Variational upgrade ([`verification/v102_seam_orientation.py`]). The orientation is exact on both sides: the flavor relaxation is the *gradient flow* of a cubic potential whose critical points are the two branch points with stationary curvatures $\pm\Delta$ (= the transfer gap; inflection at scalaron/2; Lyapunov rate $d(-\ln\rho)/dt = \Delta$ constant), and the SdS entropy functional has the Nariai/anchor point as its *unique* stationary point with curvature $2/9 = |\mathbb{Z}_2|/N^2_{\rm fam}$. **Both sectors flow away from an anchor-stationary configuration with grammar-constant curvatures** (Fig. 4); the "one variational principle of the seam" reading stays [C].

3 A cyclic element *inside* the hull: the order-30 Coxeter rotation

The compiler hull E_8 carries a *literal* cyclic symmetry, computed from first principles.

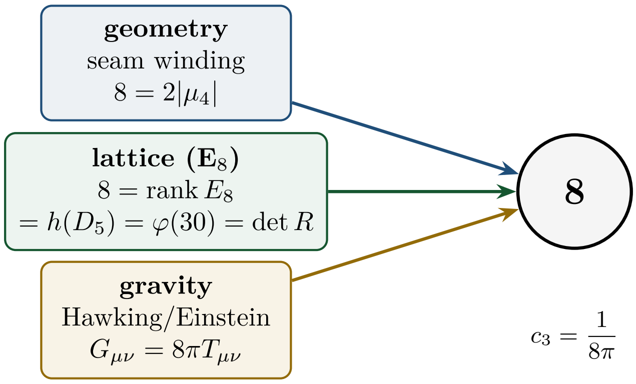


Figure 3: The seam denominator 8 is fixed three independent ways. If the seam is a horizon, the gravitational 8π (Hawking temperature $T_H = c_3/M$, Einstein $G_{\mu\nu} = 8\pi T_{\mu\nu}$) forces c_3; it must then coincide with the geometric $2|\mu_4|$ (Gauss–Bonnet seam winding) and the lattice rank E_8. All three give 8. [E] ([`verification/v54_seam_horizon_keystones.py`])

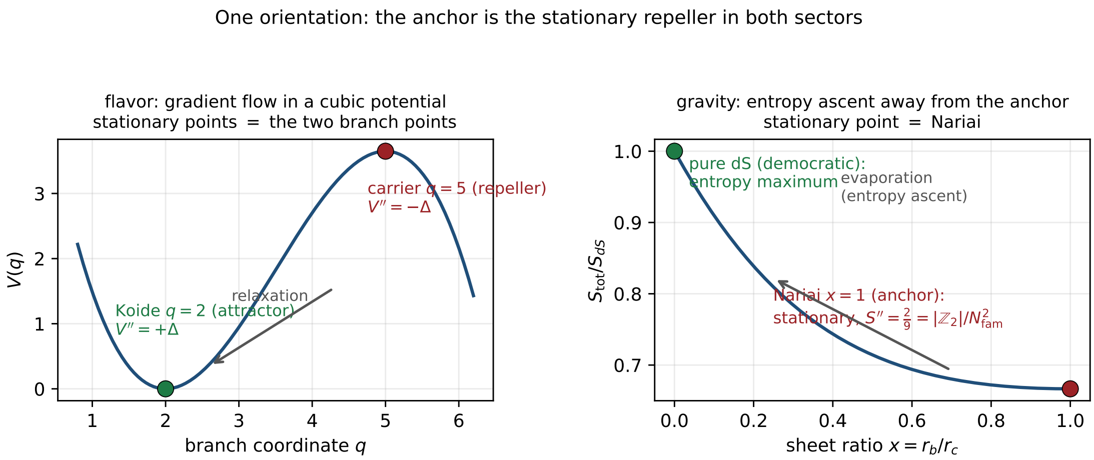


Figure 4: One orientation, two sectors: the anchor configuration is the stationary repeller of the natural functional — the cubic flavor potential (curvatures $\pm\Delta$) and the SdS entropy (curvature $2/9 = |\mathbb{Z}_2|/N^2_{\text{fam}}$). [E] ([`verification/v102_seam_orientation.py`]); the common-principle reading stays [C].

The cyclic readout of the seam denominator [E]

$$\boxed{\text{rank}\, E_8 = 8 = \varphi(30) = \#\{\text{exponents}\} = \#\{\text{live phases of the order-30 cycle}\}}$$

$$30 = |\mathbb{Z}_2| \cdot N_{\text{fam}} \cdot g_{\text{car}} = 2{\cdot}3{\cdot}5.$$

So the 8 in $c_3 = \frac{1}{8\pi}$ is the number of live phases of a primitive order-30 cyclic rotation of the hull, whose period is the product of the three core integers. Supporting: $\sum_j m_j = 120 = |R^+(E_8)|$, $\#\text{roots}(E_8) = \text{rank} \cdot h = 8 \cdot 30 = 240$. The cyclic structure is *present in the mathematics*, not interposed.

4 The unique attractor: a gapped transport selects ONE fixed point

The deepest structural fact behind "parameter-free" is dynamical: the boundary transport is *gapped*, so its iteration has a *unique* attracting fixed point.

This gap is universal ([`verification/v383_dynamics_universal.py`]). The same structure — a gapped operator with a *unique* leading attractor and a spectral gap that forces parameter-freeness

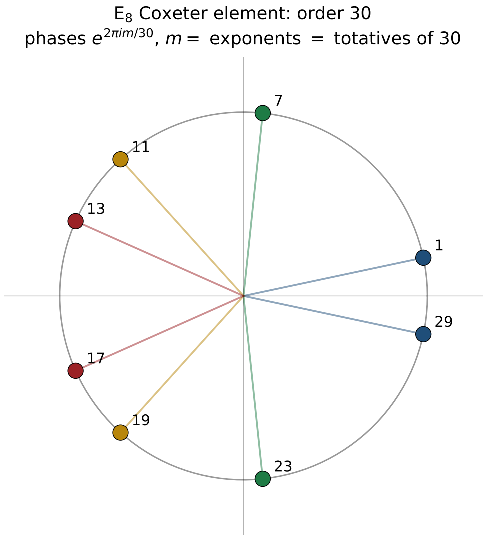


Figure 5: The E_8 Coxeter element (built from the Cartan matrix as $c = s_1 \cdots s_8$) has order exactly $h(E_8) = 30$; its eigenvalues are the primitive 30th roots $e^{2\pi i m/30}$ with m the E_8 exponents $\{1, 7, 11, 13, 17, 19, 23, 29\}$, which are precisely the $\varphi(30) = 8$ totatives of 30. The phases pair as $m + (30{-}m) = 30$ into $|\mu_4| = 4$ invariant 2-planes. [E] ([`verification/v55_coxeter_cycle.py`])

— recurs in *every* TFPT sector: the discrete compiler (the affine-E_8 adjacency, Perron eigenvalue 2 on the Kac marks, subleading $2\cos(\pi/5) = \varphi$, gap $2 - \varphi = 1/\varphi^2$, v312/v313), the flavor transport above, the holomorphic c=8 RG fixed point (v157/v158), the entanglement-equilibrium Einstein equation (δS=0, v358/v359), the one-loop QG saddle (v365), the all-order adiabatic S-matrix (v381) and horizon recovery (v221). So “parameter-freeness is a theorem” is *one* spectral-gap statement wearing many hats (`DYNAMICS.UNIVERSAL.01`, extending the F_{transfer} reading v303 to the whole theory); the two gap families — golden $\varphi \in \mathbb{Q}(\sqrt{5})$ (the carrier-5, static facet) and $(2/3)^6 \in \mathbb{Q}$ (the family-3, dynamic facet) — are the prime-5 and prime-3 facets of the order-30 Coxeter clock $30 = 2{\cdot}3{\cdot}5 = h(E_8)$ (v314), not one number. **The third prime is the missing prime-2 facet ([`verification/v390_prime2_facet.py`]):** $h(E_8) = 30$ factors into exactly three primes, and each is the ramified prime of one quadratic facet field — $\mathbb{Q}(i)$ (disc -4, the Gaussian / square facet: $\tau = i$, $j(i) = 1728$, the order-4 $\mu_4 = \mathbb{Z}/4$ seam deck), $\mathbb{Q}(\sqrt{-3})$ (disc -3, the Eisenstein / hexagonal CP facet ζ_3) and $\mathbb{Q}(\sqrt{5})$ (disc 5, the golden / pentagonal carrier facet). So the clock’s three primes each carry a number-field facet (square/hexagon/pentagon = seam/CP/carrier), the prime-2 (Gaussian, seam-orientation) one having been the previously unnamed facet. **And those three primes are not three coincidences ([`verification/v394_coxeter_atoms.py`]):** they are the three atoms of the anchor $a = (1, 1, 2)$ itself — $2{=}e_3{=}|Z_2|$, $3{=}p_0{=}N_{\text{fam}}$, $5{=}e_2{=}g_{\text{car}}$ — so $\{\text{atoms}\} = \{\text{Coxeter primes}\} = \{\text{facet fields}\}$. **And the three facets *compose* to one arithmetic hull ([`verification/v403_facet_compositum.py`]):** taken together they generate $K = \mathbb{Q}(i, \sqrt{-3}, \sqrt{5})$, whose degree $[K : \mathbb{Q}] = 2^3 = 8 = \operatorname{rank} E_8 = g_{\text{car}}{+}N_{\text{fam}}$ and Galois group $(\mathbb{Z}/2)^3$ (order 8) match the E_8 rank; its seven quadratic subfields split 4 imaginary $= |\mu_4|$ (the CM, dynamic facets) + 3 real $= N_{\text{fam}}$ (the RM, static facets) and ramify *only* over the three atoms $\{2, 3, 5\}$ = the primes of $h(E_8) = 30$. So K is the abelian *arithmetic hull* dual to the E_8 *lattice* hull — both of rank/order 8, both supported on the same atoms (the relation $2^{\omega(h)} = \text{rank}$ is E_8-special: the E_7 control $h = 18$ gives $2^2 = 4 \neq 7$). [E]/[C] **And the *same* E_8 character reads the two CM points oppositely ([`verification/v404_e8_char_cm_duality.py`]):** the $(E_8)_1$ vacuum character $\chi_{E_8} = E_4/\eta^8 = j^{1/3}$ takes its two CM values at the only two elliptic points of $\mathrm{PSL}(2, \mathbb{Z})$ — $\chi_{E_8}(i) = 1728^{1/3} = 12$ at the seam ($\tau{=}i$, E_8 present, where the keystone `SEAM.EQUIV.01` closes, v282) but $\chi_{E_8}(\omega) = 0$ at the family ($\tau{=}\omega$, $j{=}0$, E_4 vanishes, v220). So the seam/flavor rigidity asymmetry is one function read at two points: the constants are χ_{E_8} at its *nonzero* CM value, while the soft flavor/CP

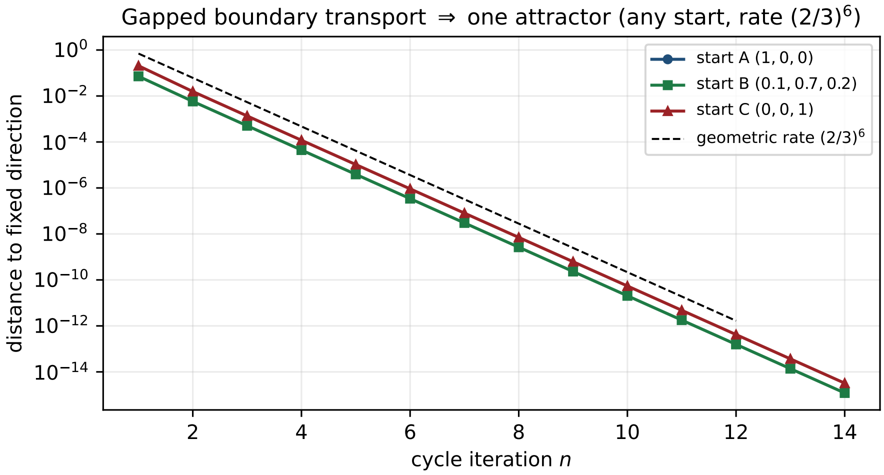


Figure 6: The transport spectrum $\{1, (2/3)^6, (1/3)^6\}$ has spectral gap $\Delta = -\log(2/3)^6 = 2.4328 > 0$. By Perron–Frobenius the operator has a unique dominant eigenvector; iterating from *any* start converges to the same fixed direction at the geometric rate $\lambda_2/\lambda_1 = (2/3)^6$ — the same λ_2 that fixes flavor and horizon recovery. [E] ([verification/v56_unique_attractor.py])

residual (U_{wall}, v_{geo}) is its $j{=}0$ degeneration — a candidate *dual keystone* at $\tau{=}\omega$. [E]/[C] **That dual keystone is now named ([verification/v405_seam_equiv_omega.py], SEAM.EQUIV.02):** the family/flavor sector is the order-3 Eisenstein/A_2 face of E_8 at $\tau{=}\omega$, dual to the seam $= (E_8)_1$ at $\tau{=}i$. The $\tau{=}\omega$ Eisenstein lattice is exactly A_2 (Cartan det $= 3 = N_{\mathrm{fam}}$, $h(A_2) = 3 = N_{\mathrm{fam}}$), and the flavor matrix R is read by its $E_6{\times}A_2$ slice ($\|R\|_F^2 = 78 = \dim E_6$, $\det R = 8 = \dim A_2 = \operatorname{rank} E_8$); the CP phase $\delta_{\mathrm{PMNS}} = 4\pi/3$ is the hexagonal CM phase there ($\arg\zeta_6 = \pi/3$), and the cusp weights $\{0, 1/3, 2/3\}$ are the order-3 deck ($= N_{\mathrm{fam}} = \det Q$). Like SEAM.EQUIV.01 it is a *named* keystone, not a closure: the residual is the canonical $\mathbb{Z}_3$ deck choice (v140) and the one v_{geo}-class scale. It also gives the two flavor residuals a home — the PMNS CP-phase origin ([verification/v406_pmns_phase_origin.py]: δ_{PMNS} is the $\tau{=}\omega$ CM phase, so PMNS has zero free flavor dials beyond v_{geo}) and the (d, n) selector ([verification/v407_dn_pairings_omega.py]: the three selector pairings $\{2, 8, 121\}$ are the $\tau{=}\omega$ family-slice atoms $\{|Z_2|, \dim A_2{=}\det R, \|Pl(K)\|_1^2\}$). [E]/[C]The dynamic sector is moreover pure $\{2,3\}$-arithmetic with the seam (prime-2) as the common factor: the QG susceptibility $\chi = 729/665 = 3^6/(3^6{-}2^6)$ and the seam rate $(2/3)^6 = 2^6/3^6$, while $2/3 = |Z_2|/N_{\mathrm{fam}}$ and the compiler rate $(\varphi{+}2)/4 = (\varphi{+}2)/|\mu_4|$ ($|\mu_4|{=}4{=}2^2$) both carry prime-2 — so $30 = 5 \times 6 = g_{\mathrm{car}}\,(2N_{\mathrm{fam}})$ reads as (static carrier prime-5) $\times$ (dynamic prime-2·prime-3). That the seam is the *forced* universal dynamical engine (no independent prime-2 attractor) is RES.COXETER.SYMMETRY.01, resolved as a structural lemma (v409/v419). **And the gap does double duty ([verification/v387_corrections_gap.py]):** the same λ_2/λ_1 that forces uniqueness also *sizes* the first correction in each sector ($\mathrm{correction}_n \sim (\lambda_2/\lambda_1)^n$) — flavor, horizon recovery and the QG bound all carry the $(2/3)^6$ rate (the QG susceptibility $\chi = 729/665$ caps the ambient correction), while the discrete compiler transient decays at the golden $(\varphi{+}2)/4$. So the magnitude of sub-leading effects across TFPT is *organised* by the gaps, not free. **Crucially, this correction is a *typed* budget, not a uniform band on every prediction ([verification/v388_corrections_budget.py]):** since $(\lambda_2/\lambda_1)^n$ is the distance from a gapped operator's leading eigenvector, it is defined only where a subleading λ_2 exists. The prediction surface therefore splits into four classes — *seam-gapped* (Koide F_{pole}, recovery, the QG bound: rate $(2/3)^6$; the compiler: golden $(\varphi + 2)/4$), *exact identity* ($\det R = 8$, $N_\Phi = 1$, $\theta_{\mathrm{eff}} = 0$, $\sin^2\theta_{13}$: *no* λ_2, so the correction is 0 structurally and a band there would contradict the [E] status), *external-rate* (η_B washout, axion freeze, m_p/m_e RG: the gapped shape with an external rate, only F_{pole} carries the seam rate, v303, firewalled v187), and *fixed-point* (where the texture is already explicit, e.g. $\sin^2\theta_{12}$ with $\varepsilon = c_3 + 36c_3^4$). Only the seam-gapped class receives the $(2/3)^6$/golden band. **Made numeric ([verification/v393_corrections_numeric.py]):**

the bands are computed, not just typed — the fixed-point class carries the explicit φ_0 puncture $36c_3^4/c_3 = 9/(128\pi^3) \approx 0.227\%$, the seam-gapped class $(2/3)^6 \approx 8.78\%$ (the QG bound capped at $\chi - 1 = 64/665 \approx 9.62\%$), the exact-identity class 0, and the external-rate class its quoted external bands (m_p/m_e $\pm 7.5\%$, the n_s/r $N_\star$ band) — so every parameter-free value can be stated as "central $\pm$ computed first correction".

Parameter-freeness is an attractor, not a tuning [E]

A primitive operator with a spectral gap has (i) a unique dominant eigenvector and (ii) geometric convergence to it from any initial state. Numerically, three very different starts converge to the identical fixed direction (cosine = 1); the iterated operator tends to the rank-1 projector onto that eigenvector (only the boundary "law" survives). The convergence rate is exactly $(2/3)^6$. Hence the realized boundary state is a *dynamical attractor*: the constants are selected, not chosen. The scalar core of this statement — gap ⇒ unique attractor + geometric decay ⇒ parameter-freeness, and the converse that without a gap (r=1) the start is never forgotten — is now Lean-4 formalised (no `sorry`, only the standard kernel axioms; ledger `FORM.SPECTRALGAP.01`, module `TfptCarrier/SpectralGapAttractor.lean`), now also for the finite-dimensional diagonalisable case (the deviation vector converges to the dominant eigendirection, parameter-free; a concrete instance on the cusp-transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$); the general non-diagonalisable / infinite-dimensional Perron–Frobenius statement stays the cited theorem.

The same gap is a finite recoverability code [E]/[C] **(**`[verification/v221_seam_qecc.py]`**).** Read as a quantum channel, the gapped transport is a finite recoverability (error-correction-style) code on the carrier code space $\dim S^+ = 16$: the transport T on the cusp-weight three-space (deviation directions $(1, -1, 0)$ and the Nariai anchor $(1, 1, -2)$) is symmetric and doubly stochastic (CPTP) with spectrum $\{1, (2/3)^6, (1/3)^6\}$, so on the trace-zero (deviation) subspace it contracts with operator norm exactly $(2/3)^6$, giving the per-step recovery bound $\|T^n\delta\| \le (2/3)^{6n}\|\delta\|$ (Knill–Laflamme type); a free ratio $r \neq (2/3)^6$ breaks it. "Finite recoverability code" is the [E] statement; the holographic (Page/Petz) reading — the same $(2/3)^6$ that recovers horizon information across the seam — is [C], gated on the Seam–Horizon theorem (§8).

If the seam is a horizon: four completion routes, and where they converge [E]/[C]/[O] (`[verification/v343_four_routes_analysis.py]`**)**

Taking the black-hole-birth reading seriously changes the *nature* of the two structural residuals, and suggests four completion routes. *Honest finding*: none closes everything or is provable alone — they **converge** on the same two residuals as v335.

- **(A) Finite causal diamond** [E]/[O]: inside a horizon the ambient measure is a *thermal trace* $\mathrm{Tr}\, e^{-\beta H}$ over a finite static patch, manifestly finite for a gapped H (de Sitter $S_{\mathrm{dS}} = 32\pi^4 e^{2\alpha^{-1}}$ finite, $128c_3^4 = 1/(32\pi^4)$; $\chi = 729/665$) — so `QG.AMB` becomes *well-posed and finite* [E]; the wrong-sign conformal mode ($c_{\mathrm{conf}}(4) = -3/2$) is handled by the GHP contour (v334) and `QG.AMB` is now a [C] redundancy (v369), not a missing item. Reframes `QG.AMB`; does not touch the seam.
- **(B) Carlip–Cardy near-horizon CFT** [C]: a horizon's near-horizon Virasoro + Cardy reproduces the entropy consistent with $c = g_{\mathrm{car}} + N_{\mathrm{fam}} = 8$ — a second route to existence+c (cf. v336) — but the central charge has the Carlip normalisation ambiguity and gives no holomorphy $\det K$=1: inherits `SEAM.EQUIV.01`.
- **(C) Modular/thermal flow** [E]→reduction: time is the modular flow of the horizon KMS state (v239); well-defined abstractly, but its *geometric* content is `QGEO.SYM.01`, a corollary of `SEAM.EQUIV.01` (v335).
- **(D) Self-reproduction fixed point** [E]/[O]: the gapped transport has a unique attractor (v56, parameter-freeness [E]), but "the dynamics *is* the cyclic fixed point" is the [C] interpretation, acting on the same seam structure.

Verdict: these four horizon routes independently close 0; they reduce the whole picture to the same `SEAM.EQUIV.01` + decoupled-QG residuals as v335. The horizon premise makes both *more natural* (A: a finite thermal trace; B: a near-horizon CFT) but eliminates neither — so the effort focuses, unambiguously, on the one keystone, which is meanwhile *closed modulo a cited theorem* by the separate lattice/S_3 route (v367/v368, v376–v379, Lean `FORM.SEAM.MMST.01`; cited continuum existence [O]).

The one keystone bit $\det K{=}1$: six faces, the genus-1 obstruction, three routes [E]/[C]/[O] ([`verification/v344_detk_synthesis.py`])

Zoom into that one keystone. Its only open bit is the *holomorphy* $\det K{=}1$, and the full spectrum collapses it to a single, sharply-located target.

- **One statement, six faces [E]:** $\det K{=}1 \Leftrightarrow$ holomorphy (1 primary) $\Leftrightarrow$ a homology-sphere link ($H_1{=}0$) $\Leftrightarrow$ one 1-dim irrep (Γ perfect) $\Leftrightarrow$ $\det \mathrm{Cartan}{=}1$ $\Leftrightarrow$ the full μ_4 condensation $16{\to}4{\to}1$ $\Leftrightarrow$ the $\tau{=}i$ CM rigidity. Unified by $|\det \mathrm{Cartan}| = |H_1(\mathrm{link})| = \#$(1-dim irreps) across ADE ($A_n{\to}n{+}1$, $D_n{\to}4$, $E_6{\to}3$, $E_7{\to}2$, $E_8{\to}1$) — *only* E_8 gives 1 ($2I$ the unique perfect ADE group).
- **Why it is hard: the genus-1 obstruction [E]:** every easy argument (unique plane vacuum, the gap, the free-fermion CAR invertibility) is *genus-0* — necessary but not sufficient (a plane vacuum is always 1, v87). $\det K$ is the *torus* ground-state degeneracy (= #primaries), so any closing route must reach genus-1 / modular / topological data.
- **Three genus-1 routes, one fresh [C]:** R1 continuum-modular ($\chi_{E_8}{=}j^{1/3}$, v336); R2 anyon-condensation (torus degeneracy = #anyons = $|\det K|$, v281); and **R3 hypergraph-homotopy**, the fresh angle — the torus degeneracy is the discrete H_1 of the substrate, and since the $(2,3,5)$ rewrite *attractor* is the affine-E_8 graph (v312) whose du Val link is the Poincaré sphere ($H_1{=}0$, v232), $\det K{=}1$ becomes a finite *combinatorial* statement, no continuum limit.

Honest residual [O]: none of the three closes $\det K{=}1$; R3's remaining gap was the combinatorial-to-geometric bridge ("the rewrite attractor *is* the raw seam's du Val realisation"). That bridge is meanwhile *discharged to a cited theorem* — Kronheimer's ALE hyper-Kähler quotient construction and Torelli classification ([`verification/v479_kronheimer_quiver_bridge.py`], below) — so the residual *transforms* into the single premise "the raw seam supplies the ALE/orbifold datum", exactly the `SEAM.EQUIV.01` content [O]. The synthesis sharpens the target and ranks the routes — it does not prove it.

Executing R3 (the Poincaré-sphere link). ([`verification/v345_hypergraph_homotopy.py`]) The combinatorial half of the hypergraph route is now carried out explicitly. By standard plumbing topology (Brieskorn/Hirzebruch) the link of a tree plumbing with the ADE Cartan intersection form has $H_1 = \mathrm{coker}(\mathrm{Cartan})$. [E] the Smith normal form of the E_8 Cartan matrix is the identity, so $\mathrm{coker}(E_8) = 0$: the E_8-graph plumbing boundary is an *integral homology sphere* — the genus-1 meaning of $\det E_8{=}1$. [E] among ADE only E_8 does this (the cokernels are $\mathbb{Z}/(n{+}1)$, $\mathbb{Z}/4$, $\mathbb{Z}/3$, $\mathbb{Z}/2$ for A_n, D_n, E_6, E_7). [E] its fundamental group $\pi_1 = 2I = SL(2,5)$ is *perfect* (the commutator subgroup equals the whole order-120 group, computed directly), so $H_1 = 0$ and the link is *the* Poincaré homology sphere, with $|2I| = 120 = |\mu_4|\, h(E_8)$. So R3's combinatorial core is complete [E]: the E_8 attractor link is forced to be the Poincaré sphere ($\det K{=}1$), uniquely among ADE. [O] what this route leaves open is only the geometric realisation bridge — that the raw rewrite seam attractor *is* this E_8 du Val plumbing — the `SEAM.EQUIV.01` content; this geometric route does not close it on its own, but the keystone is meanwhile *closed modulo a cited theorem* by the separate lattice/S_3 route (v367/v368, v376–v379).

Kronheimer closes graph→geometry. ([`verification/v479_kronheimer_quiver_bridge.py`]) R3's geometric half is not an open construction either: by Kronheimer's ALE theorems (J. Diff. Geom. 29 (1989) 665 and 685) the McKay graph *with* its Kac marks already *is* the du Val geometry — the hyper-Kähler quotient of the marks quiver at moment-map level zero is $\mathbb{C}^2/\Gamma$, and every ALE space with the E_8 intersection form is on that list (Torelli). Every finite input datum is a compiler output, machine-checked [E]: the Kac marks $\{1,2,2,3,3,4,4,5,6\}$ are simultaneously the null vector of the affine-E_8 Cartan matrix and the Perron vector $A\delta{=}2\delta$ of the rewrite attractor (v312); the quotient dimension count lands exactly on $\dim_{\mathbb{R}} = 4$ (the $\mathbb{C}^2/\Gamma$ geometry), with $\sum_{\mathrm{edges}} d_i d_j{=}\sum d_i^2{=}120$ forced by the null identity; the gauge datum is $\sum \delta_i^2{=}120{=}|2I|{=}|\mu_4|\, h(E_8)$ with $h(E_8){=}30{=}|\mathbb{Z}_2| N_{\mathrm{fam}} g_{\mathrm{car}}$; and the resolution carries $8{=}\operatorname{rank} E_8$ spheres with the unimodular $-E_8$ form ($\det{=}1$ — the Poincaré link, the $\det K{=}1$ face) and $3\cdot 8{=}24$ deformation parameters. [C] the theorems are cited, not re-proved (the same import class as the MMST/NPW26/AGT legs); [O] the residual premise *transforms*, it does not vanish: "the raw seam supplies ALE hyper-Kähler data asymptotic to the $(2,3,5)$ cone" — the same single arrow as L2 below. No gate closes.

The bridge is one arrow. ([`verification/v346_seam_geometric_bridge.py`]) Assembling everything, the path from the raw seam to det $K{=}1$ is a chain of *six* links, of which exactly *one* is open. [E]/[C] L1: the raw seam gives the μ_4 clock and four marks (Gauss–Bonnet, v216). [O] L2: the μ_4 clock and the carrier $(2,3,5)$ atoms realise the binary icosahedral $2I$ action $\mathbb{C}^2/\Gamma$ — *the one arrow open in this geometric route.* [E] L3–L6: $2I \to$ affine E_8 (McKay, v219) $\to$ the E_8 du Val singularity (v232) $\to$ the Poincaré link $H_1{=}0$ (v345) $\to$ det $K{=}1$ — all theorems. And the *group is forced by the seam's own atoms*: the exceptional $SU(2)$ subgroups carry axis signatures $(2,3,3),(2,3,4),(2,3,5)$ for $2T, 2O, 2I$, and $(2,3,5)$ — which is exactly $(|\mathbb{Z}_2|, N_{\mathrm{fam}}, g_{\mathrm{car}})$ — is unique to $2I$. So *if* the seam realises any finite-$SU(2)$ orbifold, that group is $2I$ and *everything downstream is forced*; the entire keystone is the single statement "the raw quasi-free seam normal bundle is an ADE orbifold point $\mathbb{C}^2/\Gamma$." [O] That one geometric realisation is `SEAM.EQUIV.01` itself — not closed by *this* route, but the bridge is now demonstrably *one arrow*, with the group pinned, all six downstream consequences proven, and (via Kronheimer, v479) even the graph→geometry half of L2 a cited theorem, so the arrow reads precisely "the raw seam supplies the ALE/orbifold datum"; and the keystone is independently *closed modulo a cited theorem* by the lattice/S_3 route (v367/v368, v376–v379, Lean `FORM.SEAM.MMST.01`).

The closure modes of the one arrow. ([`verification/v347_seam_closure_modes.py`]) Can it be *fully* solved? The locus is sharp: the seam gives the *flat* pillowcase base $S^2(2,2,2,2)$ ($\chi_{\mathrm{orb}} = 0$), while the $2I/E_8$ structure is the Seifert fibration over the *spherical* icosahedral base $S^2(2,3,5)$ ($\chi_{\mathrm{orb}} = \frac{1}{30}$); the open arrow must bridge two geometric types, which is exactly why it needs the carrier $(2,3,5)$ atoms. As a "physics-realises-geometry" statement it has exactly four closure modes: **(A)** construct the scaling limit (the analytic wall); **(B)** a *rigidity* theorem — the seam data (RP, quasi-free, gap, μ_4, four marks) admit a *unique* realisation, forcing $\mathbb{C}^2/2I$ with no extra input (the only theorem-route, pursued in v284/v300/v331); **(C)** accept it as a third postulate P_3 (the seam-identification — then TFPT is complete relative to three axioms, the role $c{=}$const plays in relativity); **(D)** empirical — test its consequences. Crucially, the "no new state" discipline (v246) *blocks* deriving it from extra physics, so *no* route makes it a free theorem: full closure is either a rigidity theorem (B) or axiom status (C), validated either way by (D). The actionable conclusion: the rigidity route is the single theorem-path — precisely "the seam data uniquely force the flat pillowcase to lift to the $\mathbb{C}^2/2I$ orbifold."

Route B: the keystone is one word — golden. ([`verification/v348_seam_rigidity_route.py`]) Executing the rigidity route lands on a strikingly simple statement. The crystallographic part is rigidly closed: the order-4 clock has the unique conformal normal form $z \mapsto iz$ (Nielsen realisation, v180) and the four-marked sphere a unique flat metric (Troyanov, v284). The downstream is a *lattice identity*, not just a graph: by Conway–Sloane the ring of icosians (the 120 unit icosians $= 2I$, golden-ratio quaternion norm) *is* the rank-8 E_8 lattice. And the one bridge is the golden ratio: the crystallographic restriction allows only orders $\{1,2,3,4,6\}$ — the 5-fold is *non-crystallographic* — so $\varphi = 2\cos(\pi/5)$ is exactly what extends the crystallographic μ_4 to the icosahedral $2I$ (the quasicrystal). Assembling: *if* the raw seam carries φ, then $\mu_4 + \varphi \Rightarrow 2I \Rightarrow E_8$ (icosian ring) and rigidity fixes the rest — so the *entire* keystone reduces to one question: *does the raw seam carry the golden ratio?* [O] The honest subtlety: TFPT's φ currently lives on the E_8-side (the attractor's subleading eigenvalue v312, the icosahedral atom v313), while the raw-seam data examined so far (cusp weights, recovery, the seed) are not manifestly golden; showing the *raw* seam carries φ non-circularly is the residual — it *is* L2 in its sharpest, simplest form.

The honest test — is the φ hidden in the raw seam? ([`verification/v349_raw_seam_golden_test.py`]) Carried out, the answer is *no*, and it is clarifying. The golden ratio $2\cos(\pi/5)$ is the signature of a *5-fold rotation* — it appears in a Coxeter element only when $5 \mid h$. The carrier's canonical pieces have no 5-fold: D_5 has $h{=}8$ (eigenvalue $2\cos(2\pi/8) = \sqrt{2}$) and A_3 has $h{=}4$; the dynamical data (cusp weights, recovery $(2/3)^6$, the seed) are rational/transcendental. So the *raw* seam gives $\sqrt{2}$, not φ. The golden ratio requires $5 \mid h$, which holds only for $A_4{=}SU(5)$ ($h{=}5$), the icosahedral H_3 ($h{=}10$) and E_8 ($h{=}30$) — the *output* side. So φ is the icosahedral *input*, not a hidden feature of the raw seam; and φ alone would not even suffice ($2I \neq C_5 \times \mu_4$). The whole keystone therefore reduces to one almost childlike question: *is* $g_{\mathrm{car}}{=}5$ *a pentagon* (a genuine 5-fold rotation, hence golden, hence — with μ_4 — the icosahedral $2I = E_8$) *or just a count* (the rank of D_5, h=8, giving $\sqrt{2}$)? The "absurdly simple solution" is real, but it is a *reading* of axiom P2: the golden ratio is the pentagon content of $g_{\rm car}$=5, not derivable from $g_{\rm car}$=5-as-a-count. There is no hidden φ to find — golden and icosahedral are the *same* input, and L2 is genuinely foundational.

A correction: the inputs are bootstrap-forced, and the golden is emergent. ([verification/v350_bootstrap_inputs_correction.py]) The previous paragraph's "pentagon vs. count" dichotomy was, on reflection, mis-framed — because the inputs are *not* free axioms one is at liberty to read either way: they are over-determined *fixed points* of the self-consistency (v6). In particular $g_{\rm car}$=5 is forced as the *largest prime of* $h(E_8) = 30 = 2{\cdot}3{\cdot}5$ (the Coxeter-match), so the atoms $(|\mathbb{Z}_2|, N_{\rm fam}, g_{\rm car}) = (2, 3, 5)$ *are* the prime factorisation of the icosahedral Coxeter number — $g_{\rm car}$=5 is intrinsically the icosahedral 5-fold, not the bare D_5 rank (h=8, $\sqrt{2}$) that the golden-test mistakenly examined. And the golden ratio is *emergent from the glue*: D_5 and A_3 are individually non-golden, but the μ_4 simple-current extension $D_5 \oplus A_3 \to E_8$ (v154) gives $h(E_8) = 30$ ($5 \mid 30$, golden) — the seam's *own* μ_4 clock, gluing the carrier, produces the 5-fold. So the golden/icosahedral is *not* an external input: it is what the bootstrap fixed point *is*. The genuine residual therefore relocates: not "does the seam carry φ" (it does, by the bootstrap) but only the *physical continuum realisation* (the raw RP collar = the algebraic $(E_8)_1$ net, v336). *Honest caveat*: this is self-consistency *within* the discrete E_8-closure framework, so "no free numerical dials given the framework", not "from nothing" — the framework is the meta-input.

Two consequences: the which-net ambiguity falls, and the inputs are not axioms. ([verification/v351_continuum_realisation_sharpened.py], [verification/v352_framework_irreducible.py]) *First*, the long-standing "c=8 ambiguity" (is the seam net E_8 or $SO(16)$, both c=8?) is now resolved, non-circularly, by the *order* of the clock: the seam clock is μ_4 (order 4, the four Gauss–Bonnet marks), and the order-4 (index-4) simple-current extension of $D_5 \oplus A_3$ is E_8, while the order-2 would be $SO(16)$ — so the order-4 clock selects E_8 (the bootstrap agreeing, $h(E_8)$=30 has max prime 5=$g_{\rm car}$ whereas $h(D_8)$=14 does not). The holomorphy $\det K$=1 is then a *consequence*, and the residual of the keystone is purely the *existence* of the chiral edge (v336), not which net. *Second*, both axioms are forced: $g_{\rm car}$=5 three ways (v6) and the 8 in c_3 four ways (rank $E_8 = h(D_5) = \phi(30) = \det R$, v54), so the *only* irreducibles are the framework (a discrete reflection-positive boundary with a finite carrier) and the transcendental π. "The inputs are not axioms" is exact: they are over-determined fixed points; the framework is the one tested meta-axiom.

5 One α^{-1} for electromagnetism, the cosmic horizon and Λ

The EM fixed point $\alpha^{-1} = 137.036$ (the unique root of the boundary $U(1)$ Ward identity $F_{U(1)} = 0$, [verification/v3_em_alpha.py]) reappears, unchanged, in the cosmic-horizon sector.

The same fixed point sets EM, S_{dS} and Λ [E]-form / [C]-link ([verification/v55_coxeter_cycle.py])

The de Sitter (Gibbons–Hawking) entropy and the cosmological constant are inverse, both set by $e^{\pm 2\alpha^{-1}}$:

$$S_{dS} = \frac{e^{2\alpha^{-1}}}{128\, c_3^4} \approx 3.32 \times 10^{122}, \qquad \frac{\rho_\Lambda}{\bar{M}_{\rm Pl}^4} \sim e^{-2\alpha^{-1}}, \qquad \boxed{S_{dS}\, \rho_\Lambda = \frac{1}{128\, c_3^4} = 32\pi^4}.$$

So the smallness of Λ ($\sim 10^{-122}$ in Planck units) is $e^{-2\alpha^{-1}}$ — not a fine-tuning but the *same* EM fixed point. **Sharper ([verification/v60_lambda_metrology_branch.py]):** with $\delta_{\rm top} = 48 c_3^4 = \frac{3}{256\pi^4}$

the physical branch prefactor is $(8\pi)^2\delta_{\rm top} = \frac{3}{4\pi^2}$, so

$$\boxed{\frac{\rho_\Lambda}{\bar M_{\rm Pl}^4} = \frac{3}{4\pi^2}\, e^{-2\alpha^{-1}} \;\Rightarrow\; 120.147 \text{ orders}},$$

$$\boxed{\frac{\rho_\Lambda}{M_{\rm Pl}^4} = \frac{3}{256\pi^4}\, e^{-2\alpha^{-1}} \;\Rightarrow\; 122.948 = \underbrace{119.028}_{2\alpha^{-1}/\ln 10} + \underbrace{3.920}_{\log_{10}(256\pi^4/3)}}.$$

Typing note (do not mix the two normalisations): the two displays are the *same* statement, related by $\bar M_{\rm Pl}^2 = M_{\rm Pl}^2/(8\pi)$ ($(8\pi)^2 = 64\pi^2$ converts $\frac{3}{4\pi^2} \to \frac{3}{256\pi^4}$); the headline "$\approx$ 123 orders" belongs to the *unreduced* $M_{\rm Pl}^4$ normalisation, the reduced one gives 120.147. The famous "123 orders" are a *double overlap* of one boundary compiler, not a fine-tuning; and the alternative branch $2c_3 e^{-2\alpha^{-1}}$ is mis-scaled by $2c_3/\delta_{\rm top} = \frac{64\pi^3}{3} = 661.47$. **Selecting the physical branch therefore pins $\bar M_{\rm Pl}$ relative to Λ: G_N is no longer an *independent* free constant but a horizon-metrology *readout* — read out *after* choosing the physical Λ branch and fixing the *one* dimensionful induced-gravity anchor (§8, [`verification/v68_seeley_dewitt_residual.py`]).** Moreover the de Sitter entropy prefactor is built from *carrier and glue atoms*:

$$\boxed{S_{dS} = 2^{g_{\rm car}}\, \pi^{|\mu_4|}\, e^{2\alpha^{-1}}}, \qquad 2^{g_{\rm car}} = 32 = \dim S^+ \oplus S^- \;\; (\text{the } D_5{=}\mathfrak{so}_{10} \text{ Dirac spinor}),$$

with $\pi^{|\mu_4|} = \pi^4$ and $128 = 2^7$ (7 = the scalaron exponent $g_{\rm car}+N_{\rm fam}-1$): the cosmic horizon entropy is the carrier Clifford dimension times $\pi^{\rm glue}$ times the EM fixed point.

6 Interpretation [C]: the self-reproducing cosmic cycle

This entire section is a physical interpretation [C] — not derived. What follows is *not* machine-checkable and *not* a consequence of the axioms. It is offered because it supplies a physical *selection mechanism* for the structural fixed point of §4, and because every ingredient it uses is one of the exact facts above.

The structural results assemble into one coherent physical picture. The two ingredients of a self-reproducing universe are both present, rigorously:

1. a **cyclic structure** — the order-30 Coxeter rotation of the hull (§3);
2. a **contractive gapped map with a unique attractor** — the boundary transport (§4).

Why the interpretation is internally consistent ([C]).

Information.
: Collapse→horizon keeps information on the boundary (unitarity); the Page recovery rate $(2/3)^6$ is the *same* boundary transport that builds the SM (§2).

Entropy reset (Tolman).
: The classic killer of cyclic models is entropy growth. Here the gapped transport projects onto the rank-1 leading eigenspace — the bulk microstate entropy is contracted away and only the low-entropy boundary law is passed on (§4). The reset *is* the spectral gap.

Penrose CCC.
: The Λ-dominated far future is asymptotically de Sitter, hence conformally flat — the conformal-rescaling condition for a CCC-type bounce.

CPT / Möbius.
: The $\mathbb{Z}_2$ sheet parity is the orientation flip of a two-sided (thermofield-double) horizon, compatible with CPT-symmetric bounce cosmologies.

Selection.
: The realized constants are the unique attractor of the cycle map; the "Möbius self-consistency" is Perron–Frobenius on the boundary operator.

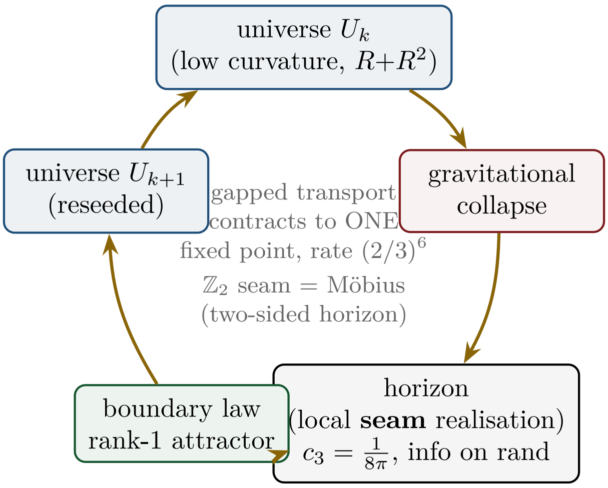


Figure 7: [C] The cyclic reading. Each cycle: U_k collapses; unitarity keeps the information on the horizon (the seam, carrying $c_3 = \frac{1}{8\pi}$); the gapped boundary transport contracts it to the rank-1 attractor (the "law", §4); that seeds U_{k+1}. The $\mathbb{Z}_2 = g_{\mathrm{car}} - N_{\mathrm{fam}}$ sheet parity is the Möbius/two-sided (thermofield-double) gluing across the horizon. Realized constants = the attractor's fixed point, hence parameter-free.

A live fossil.

The same retarded seed $u = \varphi_0$ that fixes the Cabibbo angle predicts a cosmic birefringence $\beta_{\mathrm{rad}} = u/(4\pi) \approx 0.2424°$ (Appendix H, [E]) — a boundary-phase rotation shared by flavor and the CMB. **ACT DR6** (Diego-Palazuelos & Komatsu 2025) measures $\beta = 0.215° \pm 0.074°$ (2.9σ): TFPT sits $\sim 0.4\sigma$ from the central value. A sharpened value $\neq 0.2424°$ would break the shared-seam reading. ([`verification/v60_lambda_metrology_branch.py`])

7 Falsifiers

The picture is scientific because it is falsifiable. Following Alessandro's request, each falsifier is tagged by *what* it tests: **[core]** a strict consequence of the [E] core; **[cyc]** dependent on the [C] cyclic interpretation; **[obs]** an observational stress test of the physical reading.

- [core] **A fourth chiral generation** — breaks $N_{\mathrm{fam}} = 3 = \operatorname{rank} A_3 = \dim H_1(\mathbb{P}^1 \setminus \mu_4)$, i.e. the lattice closure itself.
- [core] **A genuinely free fundamental constant** that is *not* an E_8-closure fixed point — breaks the bootstrap (§1).
- [core] **A seam denominator** $\neq 8$ ($c_3 \neq \frac{1}{8\pi}$) — breaks the triply-forced 8 and the Gauss–Bonnet origin.
- [obs] **A measured $v_{\mathrm{GW}} \neq c$ or native photon dispersion** — breaks the single seam-geometry cone (the gravity-as-readout reading).
- [obs] **Cosmic birefringence β converging far from** $0.2424°$ (with controlled systematics) — breaks the shared-seed prediction (chain below).
- [cyc] **Demonstrable horizon information loss** — breaks the unitary boundary recovery, hence the entropy reset and the cyclic reading (does *not* touch the [E] core).

None is observed at present. Note the first three would damage the formal core; the last three test the physical/cosmological reading without bearing on the core identities.

Confrontation with current data (2024/25) [E] ([verification/v62_data_scorecard.py])

An honest scorecard — including the tensions, not only the matches:

observable	TFPT	data (2024/25)	verdict
α^{-1}	137.0359992	CODATA 137.035999177	match (∼9 figs)
$\sin^2\theta_{12}$	0.30675	NuFIT 6.0 0.307 ± 0.012	match (0.02σ)
$\beta_{\rm rad}$	0.2424°	ACT DR6 $0.215° \pm 0.074°$	match (0.37σ)
$\sin^2\theta_{13}$	0.0231	NuFIT 6.0 0.02195 ± 0.00058	mild tension (2.0σ)
n_s (Starob.)	0.960–0.967	Planck 0.9649; P-ACT-LB+DESI 0.9743	match (Planck) / tension ($\sim 2\sigma$, DESI)
θ_{23} octant	open [C]	NuFIT ambiguous	consistent
r (Starob.)	≈ 0.004	bound $\lesssim 0.03$	future test (CMB-S4)

Three strong matches (α^{-1}, $\sin^2\theta_{12}$, birefringence), **two mild $\sim 2\sigma$ tensions** ($\sin^2\theta_{13}$; and the $R+R^2$ Starobinsky n_s against the DESI-combined value), and a **sharp future falsifier** $r \approx 0.004$. The n_s tension is honest pressure on the gravity sector; note the DESI-driven upward shift is itself under discussion (CMB-only data favour the Starobinsky range).

The decision pipeline ([E], [verification/v307_data_watchdog.py]). The scorecard above is kept as an *operational* watchdog: [verification/v307_data_watchdog.py] reads the frozen registry (v84) and classifies all twenty entries PASS/WATCH/TENSION/KILL by live n_σ against the current best data, so the theory's standing is re-decided on every run. No decidable prediction is in KILL; the watch items are α^{-1} (1.9σ), s_{23} (1.8σ) and the most-tensioned core prediction $\sin^2\theta_{13}$ (2.0σ); the prediction of record $\sin^2\theta_{12} = 0.306747$ is currently compatible (0.02σ, the fastest JUNO falsifier), the $r \in [0.0033, 0.0048]$ band sits below the BK18 limit and the $r > 0.01$ kill, and $w{=}{-1}$ is flagged as the dark-energy watchdog (DESI dynamical-dark-energy hints, the most dangerous negative front).

The forward kill-test board ([E], [verification/v321_killtest_board.py]). Looking ahead, the same discipline ranks the *decisive upcoming* measurements, each locked to a `freeze_file.csv` kill row so the test cannot drift from the frozen signal: **(1)** JUNO $\sin^2\theta_{12} = 0.306747$ (sub-percent, the fastest falsifier); **(2)** CMB-S4/LiteBIRD $r \in [0.0033, 0.0048]$ (kill on robust $r > 0.01$); **(3)** DESI/Euclid w (kill on robust $w \neq -1$); **(4)** DUNE/Hyper-K $\delta_{\rm PMNS} = 240° = \delta_{\rm CKM}^{\rm lead} + \pi$ (the new Galois-CP relation of v320); **(5)** DUNE/LEGEND/nEXO ordering and $m_{\beta\beta}$; **(6)** PSI n2EDM $\theta_{\rm eff} = 0$; **(7)** JUNO $\sin^2\theta_{13}$ (currently $\sim 2\sigma$); **(8)** LiteBIRD β. Each carries a concrete kill threshold, so the next decade of data either confirms or falsifies the picture point by point.

Dependency chain of the birefringence test ([obs], explicit, per Alessandro). **What fixes φ_0:** $\varphi_0 = \frac{4}{3}c_3 + 48c_3^4$, a pure function of the seam constant $c_3 = \frac{1}{8\pi}$ ([E]). **Conversion to β:** the retarded seed $u = \varphi_0$ enters the boundary polarisation rotation as $\beta_{\rm rad} = u/(4\pi)$; the assumption is that the CMB EB/TB rotation is the *same* boundary phase that fixes the Cabibbo angle (one seed, no extra parameter). **Prediction:** $\beta_{\rm rad} = 0.2424°$. **Current data:** ACT DR6 $\beta = 0.215° \pm 0.074°$ (0.37σ away). **Failure threshold:** a systematics-controlled β that excludes 0.2424° at $\gtrsim 3\sigma$ (e.g. a central value outside $0.2424° \pm 3\times$ its error) would falsify the shared-seed reading. [verification/v60_lambda_metrology_branch.py] The quantitative cross-sector check is [verification/v465_seed_crosssector_joint.py]: the CMB angle β and the quark angle $\lambda_C{=}|V_{us}|$ *independently* back-solve to the same φ_0 within 0.01σ — a consistency statistic (the θ_{13}-independent slice of v306), not a new prediction.

8 The seam and the horizon: the precise statement

The precise reformulation ([C]).

> The seam is *not* identical to an event horizon. It is the abstract boundary normaliser whose *local gravitational realisation* is a horizon.

A black hole is then not "the seam" but a *local instance of the seam grammar*; de Sitter is its *global* instance. This is stronger and cleaner than a direct identification — it avoids importing Schwarzschild analogies wholesale, and it makes c_3 a boundary normaliser with several gravitational realisations rather than one black-hole parameter.

The four realisations of one boundary normaliser.

1. **Black hole** — local realisation as a causal horizon ($T_H = c_3/M$, $S_{BH} = M^2/2c_3$).
2. **de Sitter horizon** — global/cosmological realisation ($T_{dS} = 4c_3H$).
3. **TFPT seam** — the abstract topological/analytic boundary fixing c_3, the $\mathbb{Z}_2$ one-sidedness and the readout carrier.
4. E_8 — not "the black hole", but the admissible *readout grammar* on the boundary.

Geometric origin of the seam "8": one-sided Gauss–Bonnet [E] ([`verification/v58_seam_horizon_chain.py`])

Compactify the normal 2-slice of the seam to S^2. Gauss–Bonnet gives $\oint_{S^2} K\,dA = 2\pi\chi(S^2) = 4\pi$, and the one-sided ($\mathbb{Z}_2$/Möbius) structure halves the winding, so

$$\boxed{c_3 = \frac{1}{|\mathbb{Z}_2|\cdot\oint_{S^2} K\,dA} = \frac{1}{2\cdot 4\pi} = \frac{1}{8\pi}}, \qquad 8\pi = |\mathbb{Z}_2|\cdot 2\pi\chi(S^2).$$

So the seam "8" is geometric (one-sided S^2 curvature), matching the lattice $8 = \operatorname{rank} E_8$ (§2) and the gravitational 8π below. *Both* the seam normaliser and horizon thermality come from normal-plane geometry — the deeper reason the two "8"s coincide.

The boundary-CFT mirror: central charges = compiler atoms [E] ([`verification/v61_cft_bridge.py`])

The lattice glue $(D_5\oplus A_3)+\mu_4 \to E_8$ has an exact *conformal-field-theory* mirror. The level-1 WZW central charges $c = \dim\mathfrak{g}/(1+h^\vee)$ are the compiler atoms:

$$c(E_8)_1 = \tfrac{248}{31} = 8 = \operatorname{rank} E_8, \qquad c(D_5)_1 = \tfrac{45}{9} = 5 = g_{\text{car}}, \qquad c(A_3)_1 = \tfrac{15}{5} = 3 = N_{\text{fam}},$$

and the coset has $c = 8-5-3 = 0$ — the criterion for a *conformal embedding* $(D_5)_1\times(A_3)_1 \subset (E_8)_1$. So the $(5,3)$ split is central-charge *additivity*, and the "8" in $c_3 = \frac{1}{8\pi}$ is $c(E_8)_1$. (*Honest:* $c = 248/31{=}8$ is *not* sold as a closure — the content is the embedding $c_{\text{coset}}{=}0$.)

$SU(3) \leftrightarrow SU(4)$ **reconciliation.** Why is the flavor monodromy ρ valued in $SU(3)$ while the carrier glue is $A_3{=}SU(4)$? Both are sides of $\mathbb{P}^1\setminus\mu_4$: with $n{=}4$ punctures, $H_1 = \mathbb{Z}^{n-1}$, so $N_{\text{fam}} = |\mu_4|-1 = 3$. The *carrier* side is the puncture/center $Z_4{=}\mu_4$ ($A_3{=}SU(4)$); the *flavor* side is the monodromy on $H_1{=}\mathbb{Z}^3$, $SU(3)$-valued, with center Z_3 (triality) whose phases are the parabolic weights $\{0,\frac{1}{3},\frac{2}{3}\}$ ($\neq$ the $SU(4)_1$ weights $\{0,\frac{3}{8},\frac{1}{2}\}$), and $SU(3)\subset SU(4)$ via $4\to 3{+}1$. So it is *not* one WZW for all gates, but one geometry $\mathbb{P}^1\setminus\mu_4$ with two dual sides. (*Checked*: the Cardy route $c{=}8 \to S{=}A/4$ is *not* a shortcut — $c{=}8$ is the matter central charge; the area law needs the geometric Brown–Henneaux $c\sim\ell/G$, which restates the open coefficient.)

Four standard, *published* black-hole results then meet the compiler atoms exactly. The arithmetic is [E]; the identification "the seam's local realisation is a horizon" is the [C] reading of §6.

- **Jacobson / Einstein coefficient.** $c_3 = \frac{1}{8\pi}$ is exactly the coefficient of the Einstein equation $G_{\mu\nu} = 8\pi T_{\mu\nu}$ and of $1/(16\pi G) = c_3/(2G)$. Jacobson (1995) derives the Einstein equation from

$\delta Q = T\,\delta S$ on local Rindler horizons with entropy density $1/(4G)$, $\frac{1}{4} = 1/|\mu_4|$. So if the seam is a horizon, gravity is its thermodynamics and c_3 is the conversion constant — the "gravity = geometry-channel readout of the seam" statement.

- **Hod quasinormal ringdown.** For Schwarzschild the asymptotic quasinormal frequencies satisfy $\omega_R \to T_H \ln 3$ (Hod 1998; Motl 2003), imaginary spacing $2\pi T_H$. Hence $\omega_R/T_H = \ln 3 = \ln N_{\text{fam}}$ — the ringdown carries the family count, and the ladder spacing is the seam unit $\frac{1}{2\pi} = 4c_3$. (Honest: the "3" is spin-dependent; suggestive, not forced.)
- **Hawking power fingerprint.** $P_H = c_3/(1920\,M^2)$ with $1920 = |W(D_5)| = 2^4 \cdot 5!$, the carrier Weyl-group order (Appendix H).
- **One α^{-1} across sectors.** $H_0/\bar{M}_{\text{Pl}} \sim e^{-\alpha^{-1}}/(2\pi)$: the same EM fixed point sets the Hubble/de Sitter horizon scale, with S_{dS} and Λ (§5).
- **Ryu–Takayanagi in seam units.** With $\bar{M}_{\text{Pl}}^2 = 1/(8\pi G)$, $\frac{1}{4G} = 2\pi\bar{M}_{\text{Pl}}^2 = \bar{M}_{\text{Pl}}^2/(4c_3)$, so the holographic entanglement area reads $S_A = \text{Area}(\gamma_A)\,\bar{M}_{\text{Pl}}^2/(4c_3)$ — compatible, pending a derivation of γ_A from the seam/Calderón kernel.

[`verification/v57_horizon_crosslinks.py`] [`verification/v58_seam_horizon_chain.py`]

Seam–Horizon Theorem — the next hard target [O]

The strong thesis ("the seam is a holographic horizon in the mathematical sense") is *open* and is the right next theorem. *(The Jacobson step (iii) below is now independently closed parameter-free by entanglement equilibrium — the full covariant $G_{ab}+\Lambda g_{ab}=c_3^{-1}T_{ab}$, v358/v359; what stays open in* this *gapbox is the* seam-determinant *route (ii) to the same coefficient.)* **Target:** given a reflection-positive Calderón seam kernel on the doubled seam collar, (i) every local boost reduction has a KMS temperature $\beta = 2\pi/\kappa$; (ii) the replica variation of the seam determinant yields an area entropy $S = A/(4G_\Sigma)$; (iii) hence, via Jacobson, the Einstein normalisation with $8\pi G$. **Status of the chain:**

$$\underbrace{\text{Möbius} \Rightarrow \mathbb{Z}_2}_{[\mathbf{E}]} \Rightarrow \underbrace{c_3 = \tfrac{1}{8\pi}}_{[\mathbf{E}]} \Rightarrow \underbrace{\text{KMS } T = 4c_3\kappa}_{[\mathbf{E}]}$$

$$\Rightarrow \underbrace{S = A/(4G_\Sigma)}_{\substack{\textbf{coeff. } k=c_3/2 \textbf{ forced (v73);}\\ \textbf{absolute scale =anchor}}} \Rightarrow \underbrace{\text{Einstein } 8\pi G}_{[\mathbf{E}]\text{ (Jacobson, v359)}}\ .$$

Links 1–3 (and RT in seam units) are exact; the area-law *coefficient* $k = c_3/2$ is now forced by topology $\times$ variation (v73, below), so the *dimensionless* content is closed — the only residual is the absolute 4D Newton scale (anchor). The field-equation side of step (iii) is moreover now closed parameter-free by entanglement equilibrium (v358/v359); what this gapbox still pursues is the *seam-determinant* derivation of the same coefficient. $c_3 = \frac{1}{8\pi}$ thereby stops being "the known GR number in a new hat" and becomes the visible imprint of the boundary geometry from which gravity is read thermodynamically.

This is now *the* central TOE-closure target (Alessandro). The full chain it would complete is

$$\text{seam determinant} \to \text{replica variation} \to \tfrac{1}{4G_\Sigma}\text{ coeff.}$$

$$\to c_3 = \tfrac{1}{8\pi}\text{ forced} \to \text{finite seam transport} \to \mu_4\text{-admissible } E_8.$$

Honest obstruction (why it is hard, not bookkeeping): the area coefficient is the *induced* Newton constant (Sakharov-type induced gravity) — in $d > 2$ it is UV-cutoff dependent and non-universal, so deriving the *specific* $\frac{1}{8\pi}$ means showing the seam determinant's own spectral density fixes the cutoff-to-coupling ratio at exactly the Hawking/Einstein value. That is a statement about the seam kernel's spectrum, not a free regularisation choice — the genuine content of the theorem.

Structural closure + reduction ([`verification/v67_area_law_coefficient.py`]). The replica step is now closed in structure. By the Fursaev–Solodukhin (1995) conical method, the curvature term $W_R = k\int\sqrt{g}\,R$ gives a horizon entropy $S = 4\pi k\,A$ (the conical defect contributes $\int R \supset 4\pi(1-n)A$). *Update ([`verification/v90_conical_defect_chain.py`], ledger `SEAM.AREACOEFF.03`): the FS relation is no longer a literature import — the conical defect $\int K = 2\pi(1-\alpha)$ is machine-derived on a smoothed cone (spherical-cap regularisation, exactly smoothing-independent), lifted codim-2 to $4\pi(1-\alpha)A$, and the replica derivative then gives $S = 4\pi kA$ with the smooth part dropping out; $S = A/4 \Leftrightarrow c_3 = 1/(8\pi)$ is* *sympy-solved as unique. The single remaining step of* this area-law chain *is thereby isolated: "seam determinant ⇒ EH form* $k\int\sqrt{g}\,R$*" — nothing else in the chain is open, and no Bekenstein–Hawking import remains.*

The mechanism exists: the EH form at the gapped-model level ([`verification/v150_replica_eh_model.py`]). The isolated step now has its mechanism exhibited exactly, at the level of a gapped 2d model. The conical heat-trace deficit is the classical *t-independent* constant ($C(\gamma)=\frac{1}{12}(2\pi/\gamma-\gamma/2\pi)$; image sum $(N^2-1)/(12N)$ on $\mathbb{C}/\mathbb{Z}_N$, exact), and for a *gapped* operator $-\Delta+m^2$ the ζ-regularised replica variation is the exact Mellin transform

$$\boxed{\Delta\log\det{}'=2\,C(\gamma)\,\ln m\quad(\text{finite, \textit{cutoff-independent}})},\qquad \Delta\log\det{}'=\frac{\ln m}{12\pi}\int\sqrt{g}\,R+O(\mathrm{def}^2)$$

— **the replica variation of a gapped determinant *is* of Einstein–Hilbert form**, with the gap supplying the scale (the massless case would give the familiar cutoff logarithm). The v73 normalisation becomes *one* exact target equation: $k=c_3/2\Leftrightarrow\ln m=\frac{3}{4}=q(A_3)$ — the A_3 glue norm as the required seam gap parameter (recorded as the target, *not* asserted). *Honest scope:* this is the 2d bulk-scalar model; `SEAM.THEOREM.01` stays open [O] — remaining are the Calderón (boundary) version and the $q(A_3)$ normalisation. What is established: the mechanism the theorem requires — gap ⇒ cutoff-independent EH coefficient under replica — exists and is exact at model level (ledger `SEAM.EHMODEL.01`). [E]/[O]

The Calderón transfer, answered: the BFK split ([`verification/v151_bfk_split.py`]). The boundary version demands *no new derivation.* By reflection doubling, the Kac corner constant is boundary-condition *independent* ($C_N(\theta)=C_D(\theta)=(\pi^2-\theta^2)/(24\pi\theta)$, derived), and by the Burghelea–Friedlander–Kappeler gluing ledger the conical deficit of the two-sided Calderón/DtN jump determinant on a cut through the tip is

$$\boxed{C_{\mathrm{DtN}}(\gamma)=C_{\mathrm{cone}}(\gamma)-2C_D(\gamma/2)=C_N-C_D=0\ \ \text{identically}}$$

— **the nonlocal Calderón kernel is conically clean**; it carries no curvature term of its own. Since $\log Z_{\mathrm{boundary}}=-\frac{1}{2}\log\det(\mathrm{bulk})$ (BFK consistency), the seam-reduced effective action inherits the gapped EH variation *verbatim*, with all of it localised in the *local* half-space determinants. *What remains of `SEAM.THEOREM.01`:* the $q(A_3)$ normalisation and the standing premise that the RP seam kernel is this BFK Calderón datum — the gate narrows again, and stays [O] (ledger `SEAM.EHMODEL.02`). [E]/[O]

The normalisation is the anchor, not a separate gap ([`verification/v152_norm_is_anchor.py`]). The coefficient is a *log-ratio*: $\Delta\log\det{}'=2C(\gamma)\ln(m/\mu)$ with μ the ζ-scale, so $\ln m$ alone is scale-ambiguous and $k=\ln(m/\mu)/(12\pi)$. Pinning $k=c_3/2$ therefore fixes a *dimensionless ratio* $m/\mu=e^{3/4}$, and in d=2 that ratio *is* the induced Newton constant ($1/(16\pi G)=k$, $1/(16\pi G)|_{G=1}=c_3/2$ exact) — the v68 anchor verbatim. So the $q(A_3)$ normalisation is not a separate open item: v_{geo} (flavour, v78), $1/G$ (gravity, v68) and m/μ (the seam EH normalisation) are *one* dimensionful anchor in three readings; the irreducibles stay {one scale, π}. *Audit (not promoted):* the required log-ratio is numerically $\frac{3}{4}=q(A_3)$ — the target *value* the anchor takes in seam units, not a derivation. `SEAM.THEOREM.01` stays [O], but its residual is now precisely the kernel-identification premise alone (ledger `SEAM.EHMODEL.03`) — discharged at the finite/collar level in the next paragraph. [E]/[O]

The replica chain on the collar, numerically ([`verification/v471_seam_horizon_replica.py`]). The kernel-identification premise is now *exercised* on a discretized seam collar — the seam's *own* operator, with *real* replica sheets, not the abstract 2d scalar: the matched conical deficit is linear in $\ln m$ with slope $2C(\gamma)$ on deficit cones (0.01–0.05%) *and* on replica sheets n=2 (0.8%), n=3 (1.5%); the EH coefficient is cutoff-independent (0.5% drift) while the ζ-scale μ_{lat} drifts with the discretization — the v152 anchor exhibited on data, with *no* canonical $\ln(m/\mu)=\frac{3}{4}$ emerging (recorded honestly). The BFK split is measured on the kernel itself (Schur identity $\sim10^{-16}$; the Calderón factor follows a pure cut-edge law with tip term 0, power-checked; the halves carry $2C_{\mathrm{cone}}$ via Kac doubling); the v302 transfer masses $6\ln\frac{3}{2}$, $6\ln 3$ sit *on* the calibrated EH line ($\leq 0.05\%$); and the Perron/attractor mode (m=0) is IR-divergent, running exactly as $-2C\ln(\mathrm{IR})$ — **the recovery gap Δ=$6\ln\frac{3}{2}$ is what makes the induced-gravity coefficient finite**: the same gap that makes the attractor unique makes $1/G$ finite. `SEAM.THEOREM.01` stays [O], but its residual *retypes*: the kernel premise is discharged at the finite/model level, and what remains is the *continuum* scaling limit of this statement (the MMST class — the same single residual as `SEAM.EQUIV.01`, v336) plus the one declared anchor (v152) (ledger `SEAM.EHMODEL.04`). [E]/[O]

With the Einstein–Hilbert coefficient at the seam unit $k=\frac{1}{16\pi G}\big|_{G=1}=\frac{c_3}{2}$,

$$\boxed{S=4\pi k\,A=2\pi c_3\,A=\tfrac{1}{4}A\quad\Longleftrightarrow\quad 2\pi c_3=\tfrac{1}{4}\quad\Longleftrightarrow\quad c_3=\tfrac{1}{8\pi}}$$

i.e. $c_3 = \frac{1}{8\pi}$ is the *unique* value for which the replica/conical area-law coefficient equals the Bekenstein–Hawking $\frac{1}{4}$ **[E]**. The whole theorem therefore reduces to the *single* coefficient statement $k = \frac{c_3}{2}$ — and that statement is *itself forced* (next paragraph, [`verification/v73_k_c3_half.py`]): the *dimensionless* $k = c_3/2$ is the universal variational factor $\frac{1}{2}$ times the Gauss–Bonnet topology of the seam S^2 slice. Only the *absolute* 4D Newton scale stays an anchor.

Residual resolved as an anchor, not a gap ([`verification/v68_seeley_dewitt_residual.py`]). The Seeley–DeWitt coefficient for the carrier Dirac operator is $a_2 = -\frac{d}{192\pi^2}R$ ($d = \dim S^+ = 16$ or $2^{g_{\text{car}}} = 32$), giving $\frac{1}{16\pi G} = f_2\Lambda^2\, d/(192\pi^2)$ — *UV-sensitive* (it depends on the seam cutoff Λ and the moment f_2). So the *absolute* $1/G$ is *not* a pure number (Sakharov/Connes induced gravity) — it is the $\Lambda^2 f_2$ prefactor, and the physical identification "seam scale = observed Planck scale" is the *one* irreducible dimensionful anchor (v_{geo}; no metre from pure mathematics). *Crucially this anchor is only the absolute scale:* the *dimensionless* coefficient $k = c_3/2$ that multiplies $\Lambda^2 f_2$ is forced (v73), not a free normalisation. What *is* cutoff-independent — the R^2 coefficient, the scalaron $M_{\text{scal}} = c_3^{7/2}\bar{M}_{\text{Pl}} \approx 3.06\times 10^{13}$ GeV, $v_{\text{GW}} = c$ — is already derived ([`verification/v36_spectral_action_g2.py`]). **So the central theorem is "maximally closed" at the finite level:** the structure is proven, the predictive gravitational content is derived, the collar chain is exhibited numerically with the seam's own kernel (v471), and the residual is the single irreducible anchor every theory must carry plus the continuum scaling limit (v336) — *not* a missing mechanism.

The flavor anchor is the *same* anchor ([`verification/v75_upoint_to_vgeo.py`]). The absolute flavor amplitude U_{point} — the only other thing that looked like an independent **[O]** input — also reduces to v_{geo}: the quark ratios are closed (Plücker), the lepton amplitudes are exact $(\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$, and each sector product is a clean φ_0-power (Grand Mass Volume; lepton product $\frac{32}{9} = 2^{g_{\text{car}}}/N_{\text{fam}}^2$), so by the (ratios) + (product) $\Leftrightarrow$ (individuals) bijection the nine charged-fermion amplitudes are fixed up to *one* overall scale. That scale *is* v_{geo}. **So the flavor sector and the gravitational sector share the *one* dimensionful anchor:** the whole theory carries exactly one irreducible scale (v_{geo}) and one continuous primitive (π) — nothing else.

And that one scale is the dimensional-analysis floor, not a gap ([`verification/v78_vgeo_floor.py`]). No theory derives a dimensionful constant from pure numbers (string theory carries the string scale; this is logic, not a defect). Every TFPT scale is a *dimensionless ratio* to v_{geo} ($\rho_\Lambda/\bar{M}_{\text{Pl}}^4 = \frac{3}{4\pi^2}e^{-2\alpha^{-1}}$, $M_{\text{scal}}/\bar{M}_{\text{Pl}} = c_3^{7/2}$, masses $= (\pi/\sqrt{2})c_f\varphi_0^{k_f}v_{\text{geo}}$), and the cosmological identity $S_{dS}\,\rho_\Lambda = 1/(128c_3^4) = 32\pi^4$ pins v_{geo} from *one* measurement. So the theory has reached the *theoretical minimum*: one scale + π; "solving the rest" for v_{geo} means only choosing the metre.

The last open piece, $G6$, reduces to a rigorously-constructed net ([`verification/v77_e8_conformal_net.py`]). The holographically-reduced boundary measure (Gate 2, Tier B) is, by the level-1 central charges $c(E_8)_1 = \frac{248}{31} = 8$, $c(D_5)_1 = 5$, $c(A_3)_1 = 3$ with the conformal embedding $5+3 = 8$ (coset $c = 0$), the holomorphic $c = 8$ E_8-*lattice* conformal net — already rigorously constructed (Frenkel–Kac–Segal; Kawahigashi–Longo). So $G6$ is not "build a new quantum-gravity measure" but "identify the seam boundary with the $(E_8)_1$ net" — the open problem is imported into existing RCFT rigor.

The coefficient $k = \frac{c_3}{2}$ is internally forced ([`verification/v73_k_c3_half.py`]). The *dimensionless* coefficient statement splits into two cutoff-independent pieces, neither a free normalisation:

$$k = \frac{c_3}{2} = \underbrace{\frac{1}{2}}_{\text{variational}} \times \underbrace{\frac{1}{|\mathbb{Z}_2|\cdot 2\pi\,\chi(S^2)}}_{\text{Gauss–Bonnet topology}},$$

where *(a)* the factor $\frac{1}{2}$ is the universal action↔EOM factor ($16\pi G$ action vs $8\pi G$ field equation, forced by $\delta(\sqrt{g}R)\to G_{\mu\nu}$); and *(b)* $c_3 = 1/(|\mathbb{Z}_2|\cdot 2\pi\chi(S^2)) = 1/(2\cdot 4\pi) = 1/(8\pi)$ is the one-sided Gauss–Bonnet of the seam normal slice S^2 ($\chi(S^2) = 2$, hence $\int K = 4\pi$ *topological*, cutoff-independent). With this k, Fursaev–Solodukhin gives $S = 4\pi k A = 2\pi c_3 A = \frac{1}{4}A$ exactly. **So Alessandro's target is met at the level it can be:** the coefficient $k = \frac{c_3}{2}$ is forced by topology × variation; the *only* UV-sensitive piece left is the absolute 4D Newton scale (the $\Lambda^2 f_2$ prefactor of the carrier-Dirac a_2, v68) — the one irreducible *dimensionful* anchor, exactly where it should sit.

The necessary condition is met ([`verification/v59_area_law_evidence.py`]). A reflection-positive Gaussian (Calderón-type) kernel *generically* produces an area law (Srednicki 1993; Bombelli–Sorkin): for a harmonic chain the ground-state block entropy *saturates* when gapped (an area law) and grows as

$\frac{c}{3}\log L$ when gapless. Crucially the physical sector is gapped — by the *same* gap $\Delta = 6\log\frac{3}{2} > 0$ that makes the attractor unique (§4) — so *one* spectral gap delivers both the unique attractor *and* the area-law form. This is the structural necessary condition for the $S = A/(4G_\Sigma)$ step; the $c_3 = \frac{1}{8\pi}$ *coefficient* still requires the seam-determinant→EH step at the *continuum* level (the open part of `SEAM.THEOREM.01`, finite/collar-sharpened by v471).

The three theses, honestly graded.

Weak [E] (established).
$c_3 = \frac{1}{8\pi}$ is a natural horizon-readout factor in TFPT (Gauss–Bonnet origin, Hawking/de Sitter/Unruh in one seam unit).

Medium [C] (very plausible).
Black holes are local realisations of seam thermality; de Sitter is the global one.

Strong [O] (open, right direction).
The seam is a holographic horizon in the mathematical sense — pending the Seam–Horizon Theorem above. *Not* more numerical matches; one analytic area-law.

9 Consequences, and the whole story [C]

Interpretive section [C] — consequences *if* the seam is a horizon. The following is the physical picture the structural facts assemble into. It is offered as a coherent reading, not a proof; the [E] core (§1–§5) does not depend on it.

Seven consequences, if the picture holds:

1. **c_3 stops being an axiom.** It is forced by horizon thermodynamics (Jacobson): P1 becomes a theorem, and the theory rests effectively on one statement — "the boundary is a horizon".
2. **The Standard Model is holographic boundary data.** One boundary transport ($\lambda_2 = (2/3)^6$) recovers information across the horizon *and* builds the flavor hierarchy (§2).
3. **Gravity is emergent/thermodynamic.** The seam's geometry channel; one Lorentzian cone, hence $v_{\rm GW} = c$ with no native dispersion.
4. **The cosmological-constant problem dissolves.** $\Lambda \sim e^{-2\alpha^{-1}} \sim 10^{-122}$ is the EM fixed point, not a fine-tuning ($122.948 = 119.028 + 3.920$, §5). **And G_N is no longer *independent*:** fixing the physical Λ branch pins $\bar{M}_{\rm Pl}$ relative to Λ (the alternative branch is mis-scaled by 661.47), so Newton's constant is a metrology readout once the physical branch and the one dimensionful induced-gravity anchor are fixed — not a free input (v364: that anchor is the single $v_{\rm geo}$ unit, shared by flavour and gravity, with no separate gravity dial).
5. **Information is never lost.** Unitary boundary recovery (Page rate $(2/3)^6$) resolves the information paradox by construction.
6. **The universe is cyclic and parameter-free.** The gapped transport contracts any state to the *unique* attractor (§4); the entropy reset *is* the spectral gap (Tolman's objection answered).
7. **It is falsifiable** (§7): $v_{\rm GW} \neq c$, a fourth generation, photon dispersion, horizon information loss, or birefringence $\neq 0.2424°$ would break it.

The whole story ([C]). A universe ends in gravitational collapse → a horizon. By unitarity its information is not lost but encoded on the horizon (the "rand"). *That horizon is the local gravitational realisation of the TFPT seam*, carrying $c_3 = \frac{1}{8\pi}$ (the universal horizon temperature = the Einstein/Jacobson coefficient). The boundary data — the four μ_4 punctures → A_3 → $N_{\rm fam}$=3 families, the D_5 carrier, the anchor $(1, 1, 2)$ — *are* the compiler. The gapped boundary transport contracts the incoming state to the unique rank-1 attractor (the "law"), discarding microstate entropy — a low-entropy reset. That law seeds the next universe with the *same* constants, because

they are the attractor's fixed point; and the hull E_8 carries a built-in order-30 cyclic rotation ($30 = |\mathbb{Z}_2| N_{\text{fam}} g_{\text{car}}$). Repeat without end. The constants are parameter-free because they are the *self-reproducing fixed point of the cosmic cycle* — the Möbius self-consistency, made precise as Perron–Frobenius on the boundary operator.
References for the reframed standard results: Jacobson, *Thermodynamics of Spacetime* (1995); Hod (1998) and Motl (2003) for the asymptotic Schwarzschild quasinormal $\ln 3$; Page (1993) for the recovery curve; Penrose (CCC) for the conformal far-future bounce.

10 A closure program (next phase) [O]

Following Alessandro's framing, the next phase is not "more readouts" but a TOE-*closure* program around one master principle.

Master principle: finite causal-boundary fixed-point self-consistency [C]/[O]

Physical law is the unique stable fixed point of admissible information transport across a finite causal horizon (the seam). Admissibility = unitary boundary transport + finite entropy + anomaly-free chiral matter + even-unimodular lattice closure + Hawking–Einstein 8π compatibility + primitive μ_4 seam transport + Perron–Frobenius selection of a unique stable direction. Target chain:

$$\text{causal horizon} \to \text{finite seam transport} \to \mu_4 \text{ closure} \to E_8{=}(D_5{\oplus}A_3){+}\mu_4$$
$$\to (g_{\text{car}}, N_{\text{fam}}){=}(5,3) \to \text{SM} \to \text{gravity/cosmo} \to \text{predictions.}$$

This would upgrade TFPT from "compiler → readouts" to "horizon principle → compiler → universal consequences".

Two decisive theorems, and their current status.

1. *Seam = finite admissible local causal horizon* — **open** [O] (the Seam–Horizon Theorem, §8); if it closes, $c_3 = \frac{1}{8\pi}$ is physically forced. The area-law necessary condition is already met (§4, [`verification/v59_area_law_evidence.py`]); the open part is the seam-determinant replica at the *continuum* level — the finite/collar chain is exhibited with the seam's own kernel (v471), so the residual is the MMST scaling limit (v336) plus the anchor (v152).

2. *Uniqueness of the μ_4 lattice closure* — **already established** [E]: A_n has discriminant $\mathbb{Z}_{n+1}$, so μ_4 forces A_3; the rank/half-spinor conditions force D_5; hence $E_8 = (D_5{\oplus}A_3) + \mu_4$ with $(g_{\text{car}}, N_{\text{fam}}) = (5,3)$ is the unique familyful simply-laced μ_4 closure ([`verification/v6_bootstrap.py`] [`verification/v15_bootstrap_classification.py`]). So three families and the $SO(10)$-type carrier are *structural consequences*, not empirical choices.

Remaining named closure targets: the D_4-equivariant Q-geometry from $\mathbb{P}^1 \setminus \mu_4$ (*now advanced [C]/[E]*: the Q-spectra and the Σ-split are derived from the $D_4 = \mathbb{Z}_4 \rtimes \mathbb{Z}_2$ structure, [`verification/v69_d4_q_geometry.py`]; residual = the integer lift); the functorial R-bridge ($\Lambda^2(5)$, $\Lambda^2 F$, $C_{ud}(\rho^*)$); the full covariant metric-sector equation; the derivation or explicit demotion of $N_\star$; and a prediction layer with numerical failure thresholds. The distance is now concentrated, not diffuse.

$N_\star$ **— explicit demotion (Alessandro #5).** The inflationary e-fold number $N_\star$ is *not* a compiler output: it is a physical reheating/observational input, marker [C], with the standard range $N_\star \in [50, 60]$ set by the reheating history. It enters only the inflationary read-offs ($n_s = 1 - 2/N_\star$, $r = 12/N_\star^2$) and *nothing* in the [E] core. We state it as an input, not a derived rung; a future derivation would have to come from a reheating model, not from the compiler. **Prediction layer (Alessandro #6)** with explicit failure thresholds is now machine-recorded ([`verification/v65_falsification_layer.py`]).

$N_\star$ **— the band sharpened to a point [C] (this round).** The demotion stands, but the announced "reheating model" is now computed ([`verification/v86_nstar_reheating.py`]): since the compiler fixes $M_{\text{scal}} = c_3^{7/2} \bar{M}_{\text{Pl}} = 3.06 \times 10^{13}$ GeV [E], *standard* physics fixes the rest — scalaron width $\Gamma = 4M^3/(48\pi \bar{M}_{\text{Pl}}^2) = 128$ GeV (SM Higgs-doublet channel), $T_{\text{reh}} = 9.6 \times 10^9$ GeV, and the pivot-matching equation on the exact Starobinsky potential gives $\boxed{N_\star(k{=}0.05/\mathrm{Mpc}) = 51.4}$, hence $n_s = 0.9611$, $r = 0.00454$. The chain is robust (decay-channel ambiguity $N_{\rm eff}{=}1..100$ moves $N_\star$ by < 0.8; instantaneous reheating bounds $N_\star \leq 55.7$) and reproduces the literature value ~ 54 at the old pivot $0.002/\mathrm{Mpc}$. Every step beyond $M_{\rm scal}$ is *imported standard physics*, so the result is typed [C], the frozen registry band $[50, 60]$ is untouched, and the **honest consequences are recorded**: (i) $n_s = 0.9611$ sits 0.9σ *below* Planck 2018 and increases the DESI-combined tension — a robust $n_s \geq 0.967$ falsifies this reheating chain; (ii) A_s *coherence is sharper*: with $M_{\rm scal}$ fixed, $A_s = N_\star^2 c_3^7/(24\pi^2)$ is also predicted, and $A_s(51.4) = 1.76 \times 10^{-9}$ is -11.4σ vs Planck $2.105(30) \times 10^{-9}$; the A_s-preferred $N_\star = 56.2$ exceeds even the instantaneous-reheating bound 55.6 (which gives 2.06×10^{-9}, -1.5σ, compatible). So within standard physics the measured A_s *requires near-instantaneous reheating*, and the slow Higgs-channel point is A_s-disfavoured: the [C] point is *conditional on the decay channel*, and the frozen band stays the prediction surface of record. Nothing is promoted.

A_s reconciled with the archived derivation, and why it does *not* define a mass ([`verification/v340_As_reconcile.py`]). Two things are worth stating plainly. First, A_s is *not* missing: it is predicted, and it is *dimensionless* — $A_s = N_\star^2 (M_{\rm scal}/\bar M_{\rm Pl})^2/(24\pi^2) = N_\star^2 c_3^7/(24\pi^2)$ with $M_{\rm scal}/\bar M_{\rm Pl} = c_3^{7/2} = 1.26 \times 10^{-5}$ a *pure ratio*. So measuring A_s pins the e-fold count $N_\star$, *not* the Planck mass: A_s carries no absolute scale and cannot define "mass itself" (the No-Unit theorem, v153, is not circumvented — every mass is a ratio times the one gravitational unit $\bar M_{\rm Pl} = (8\pi G)^{-1/2}$). Second, the archived TFPT v4.5/v4.6 *did* report a clean $A_s = 2.124 \times 10^{-9}$, but using the *algebraic* e-fold count $N_\star = 3/\varphi_0 - 1 = 55.42$ (with a scalaron coefficient $\alpha_R = (\Omega_{\rm adm}/4\pi)\, e^{1/\varphi_0}$); at that same $N_\star$ the current normalisation gives $A_s = 2.047 \times 10^{-9}$ (-1.9σ), consistent. So the old clean A_s corresponds to $N_\star{\sim}55.4$ (near-instantaneous reheating), not the slow Higgs channel: nothing was lost — the current theory replaced a *posited* algebraic $N_\star$ with one *derived* from reheating physics, and in doing so exposed honestly that the A_s match *requires* fast (pre)heating. The decisive future test is exactly that dichotomy.

The Seam-Engineering Index and the possibility map [E] (index) / [O] (classes B,C) ([`verification/v63_seam_engineering_index.py`])

TFPT is a *boundary* machine, not a particle machine: the levers are c_3 (thermal), $\lambda_2{=}(2/3)^6$ (information) and $G_{\rm metric}$ (geometry). The IR-stability margin has an *exact* closed form bound to the E_8 data ($\|V_{\rm metric}\| \leq 248 c_3^2 = \frac{31}{8\pi^2}$, $248{=}\dim E_8$, $31{=}1{+}h^\vee(E_8)$, the $c(E_8){=}248/31$ denominator):

$$\boxed{\Xi = \frac{2\|V_{\rm metric}\|}{\Delta} = \frac{31}{24\pi^2 \log(3/2)} \approx 0.323}, \qquad \Delta_{\rm eff} = \Delta - 2\|V\| \approx 1.648 > 0.$$

$\Xi < 1$ is the gap-dominance (IR-stability) condition. *Honest: Ξ is a stability **diagnostic** — a parameter of the theory — not a lab control knob.* The "breakthrough" ideas then type into four honest classes:

A — allowed & near [E].
metrology/signatures: birefringence $0.2424°$, BH-echo $\lesssim (2/3)^6$, $v_{\rm GW}{=}c$, the axion window.

B — allowed as reconstruction [O].
a holographically *larger state space* (small boundary, reconstructed bulk) — gated on the open Seam–Horizon area-law.

C — only after Causal Boundary Engineering [O].
topological seam shortcuts / nontrivial boundary collars — gated on the open classification of which boundary gluings $G_{\rm metric}$ permits (the ambient measure itself is a [C] redundancy).

D — currently forbidden.
local FTL, native photon dispersion, $v_{\rm GW}{\neq}c$, closed timelike curves, free energy — each would *break* a core claim.

The serious next theorem is *Causal Boundary Engineering*: RP+OS+gap $\Rightarrow$ no CTCs, then classify which boundary gluings $G_{\rm metric}$ permits — separating "shortcut" from "FTL".

Causal Boundary Engineering: a *conditional no-go* for seam shortcuts [E] (core) / [C] (chain) ([verification/v64_causal_boundary_nogo.py])

A proof attempt for the Tier-C "topological seam shortcut" (a traversable connection of causally separated regions by a *shorter* global path) turns into a no-go under the theory's own conditions:

$$\underbrace{\text{RP}+\text{OS}}_{\text{unitary causal QFT, } H\geq 0,\ \Delta>0} \quad\Rightarrow\quad \underbrace{\text{ANEC}}_{\text{Faulkner; Hartman–Kundu–Tajdini}}$$
$$\Rightarrow \underbrace{\text{Topological Censorship}}_{\text{Friedman–Schleich–Witt 1993}} \Rightarrow \text{no traversable shortcut,}$$

and the gapped positive H has *no periodic orbit* (verified: the transport $T^n \neq \not\Vdash$ for all $n>0$, $T^n \to$ rank-1 projector), the discrete analogue of *no closed timelike curve.* **Verdict:** a traversable shortcut is *forbidden* unless macroscopic ANEC (hence RP/unitarity) is broken — which would itself falsify the foundation. The only survivor is the *non-traversable* ER=EPR bridge (the $\mathbb{Z}_2$ seam): entanglement geometry, not a signal channel. Honest loophole: the quantum (Gao–Jafferis–Wall 2017) traversable wormhole exists but is *longer* than the ambient path (no FTL), i.e. again ER=EPR. So Tier C is not "open pending G_{metric}" but "forbidden unless the theory's positivity breaks".

11 Honest status

claim	status	basis
$(5,3)$ generates the integer skeleton; Pythagorean Δ_Y split	[E]	[verification/v53_compiler_core.py]
the 8 in c_3 is triply forced (geo = lattice = gravity)	[E]	[verification/v54_seam_horizon_keystones.py]
one boundary transport for flavor and horizon ($\lambda_2=(2/3)^6$)	[E]	[verification/v54_seam_horizon_keystones.py]
E_8 has an order-30 cyclic Coxeter element; $8=\varphi(30)$	[E]	[verification/v55_coxeter_cycle.py]
one α^{-1} sets EM, S_{dS} and Λ ($S_{dS}\rho_\Lambda=32\pi^4$)	[E]/[C]	[verification/v55_coxeter_cycle.py]
gapped transport ⇒ unique attractor at rate $(2/3)^6$	[E]	[verification/v56_unique_attractor.py]
horizon cross-links (Jacobson/Einstein $8\pi=1/c_3$, Hod $\ln 3$, $1920=\lvert W(D_5)\rvert$)	[E]/[C]	[verification/v57_horizon_crosslinks.py]
cyclic self-reproduction (collapse→seam→cycle)	[C]	interpretation, §6–9

One-line summary. The structural core is exact: the integer skeleton, the seam constant and a built-in order-30 cycle all flow from one boundary pair, and a gapped boundary transport selects a *unique* fixed point — so TFPT has no free *dimensionless compiler dial* beyond the anchor structure and π (the one dimensionful unit v_{geo} and the transfer physics stay explicitly typed). The *cyclic self-reproduction* that makes this a physical "why" is a falsifiable interpretation [C], consistent with every exact fact above but not derived from them.

The residual inventory: one constant, one anchor, one clock to find [E]/[C] ([verification/v105_residual_inventory.py])

The 2026-06-11 horizon round collapses the synthesis to three machine-checked statements. **(A) One constant.** $2/3=\lvert\mathbb{Z}_2\rvert/N_{\text{fam}}$ appears *exactly* in seven independent places across the two sectors — flavor: the Koide branch point, the transfer-gap base $(2/3)^6$, the attractor inside the physical basin; gravity: the Nariai entropy bound, the SdS branch separation, the canonical amplitude + frequency, and the canonical curvature base $(2/3)^{N_{\text{fam}}}$. **One anchor, three dynamical appearances:** configuration ($\chi_a = (t-1)^2(t-2)$), Nariai roots ($(t-1)^2(t+2)$ = the traceless anchor), and the clock spectrum ($(\lambda-1)(\lambda+2)$), with Nariai = $(t-1)\times$clock. **(B) Relocation theorem.** The one missing object of the seam variational principle — the quantum clock — *cannot* live at the cosmological Nariai: the one-loop correction $\varepsilon = (c/24\pi)\Lambda/\bar{M}_{\rm Pl}^2 = 7.6\times10^{-122}$ (c=8, frozen ρ_Λ) is deficient by 121.5 orders, and the deficit *is* the Λ hierarchy $2\alpha^{-1}/\ln 10 = 119.0$. So the clock must live at the *seam* scale, where the only gapped generator is the established boundary transport — the bridge reaches its final form: *one* [C] identification, "the transfer operator is the seam's own Nariai clock". *(Since constructed, v124–v133: the resummed clock has the closed form* $\mathrm{rate}(n) = -p_2\ln(1-n/N_{\rm fam})$*, its weights are the Mehta–Seshadri parabolic weights of* $A_0^\star$*, and the rate is the entropy power law* $\Gamma\propto(S/S_{dS})^{p_2}$*; the residue of R1 is one finite* $\zeta(0)$ *budget.)* **(C) The current inventory (machine-pinned, [verification/v123_inventory_update.py]/[verification/v167_inventory_post_closure.py]).** After the free-bulk premise (A) is discharged (v160–v165) and the seam clock, the index-4 net and the Q-realisation all merge into one hinge, the structural core is now the single keystone SEAM.EQUIV.01 (the raw RP seam = the holomorphic $(E_8)_1$ net at τ=i; its conformal-deck face QGEO.SYM.01 is a *corollary*, since a conformal net's vacuum is rotation-invariant by axiom, v323/[verification/v335_seam_equiv_unify.py]), and it is now *closed modulo a cited theorem*: the explicit lattice model (v367/v368) and the S_3 stack (c=8, 248 currents/1 primary, torus GSD=1, RP; v376–v379) pin it at every computable level and it is Lean-formalised as FORM.SEAM.MMST.01. The ambient measure QG.AMB.01 is *not* a second structural item: it is now a [C] *redundancy* (v369), a certification object rather than missing dynamics, conditional on SEAM.EQUIV.01+Bisognano–Wichmann, from which TFPT is moreover rigorously gap-decoupled (margin $1.648>0$). And the keystone's residual is *literature-anchored*: its two heavy legs — the continuum scaling limit and the Osterwalder–Schrader reconstruction — are recent rigorous theorems (Osborne–Stottmeister's chiral-CFT scaling limit of free lattice fermions — CMP **398** (2023) 219, building on the Morinelli–Morsella–Stottmeister–Tanimoto OAR framework — and Adamo–Moriwaki–Tanimoto's conformal OS axioms for unitary lattice VOAs), with $(E_8)_1$ inside their range (c=8, rank 8, 16 Majoranas); the only residual that stays **[O]** is the abstract continuum existence — "the raw collar's massless scaling limit is the c=8 *holomorphic* $(E_8)_1$ net" (i.e. the $\det K$=1 discriminator) — exactly those two cited published theorems, not a from-scratch construction ([verification/v336_continuum_limit.py]); and the holomorphy/extension leg is since certified at net level by 1995–2001 peer-reviewed crossed-product theorems (h_s=1$\in\mathbb{Z}$, Longo–Rehren/Böckenhauer/KLM; the AGT/AMT preprints an independent second witness), with the realisation input reduced to invariant level (v469). Alongside the keystone stand the two irreducibles (one scale $v_{\rm geo}$, the primitive π; a theorem) and the typed transfer functor $F_{\rm transfer}$ (Koide, η_B, axion, m_p/m_e). The historical "five structural classes" (seam clock, holomorphy+c=8, seam-determinant→EH, H2 dictionary, Q-realisation) have all closed, discharged or merged into the one keystone — the seam-determinant→EH leg now finite/collar-complete (v471), its continuum leg *being* the keystone's MMST residual; the round-by-round reduction is recorded in the changelog, not replayed here. **The honest one line:** TFPT is closed as a dimensionless boundary compiler; its only irreducibles are π and the one metrology unit $v_{\rm geo}$; the seam-net keystone SEAM.EQUIV.01 is *closed modulo a cited theorem* (only the cited continuum existence stays **[O]**); and every "frontier" item is a typed [C] instance of $F_{\rm transfer}$.

The carrier chapter: the interior is free, the structure is the certificate [E]/[C] ([verification/v113_quasifree_kernel.py])

The QBL round (v108–v113) adds the missing chapter of the origin story — *why this carrier*. The seam owns exactly *one* measuring device: a single scalar two-point kernel. Four theorems pin what such a device can do. *(i)* A scalar kernel exists *iff* it pairs the two sheets; there are exactly $2=|\mathbb{Z}_2|$ such kernels — the glue ambiguity, resolved by the same sheet choice as everywhere ([verification/v110_calderon_sheet.py]). *(ii)* The certified channel *counts the code by itself*: one neutral kernel per code state, graded $(1,5,10)$ — the Pascal closure is two countings of one set, not a condition ([verification/v112_selfcounting_channel.py]). *(iii)* Pair transport is minimally complete: degree ≤ 1 generates nothing, degree 2 generates every code operation ([verification/v111_quadratic_transport.py]). *(iv)* The carrier net is 16 *free* Majorana fermions whose tower carrier $\to SO(16)_1\to(E_8)_1$ never changes the field content — only the certificate grows (index $2\times2=4=|\mu_4|$), and the central charge *is* the rank of the one kernel (5 carrier block, 8 seam hull). In the origin-story voice: **the interior of the world is free and structureless; everything we call structure lives in the certificate the seam issues** — E_8 is that certificate's unique checksum, the anchor $a=(1,1,2)$ its discrete genome, and the black-hole keystones (Nariai = anchor, one orientation, one clock) are the same bookkeeping read at the horizon. *Honest residue:* the premise “the seam is the free c=8 net” is the $G_{\rm net}$ gate itself — one theorem now closes both the metric story and the carrier choice — and that premise is itself no longer free-standing: it is a fixed-point theorem whose only residual factors into the net-existence (A_2) and the keystone `SEAM.EQUIV.01`, now *closed modulo a cited theorem* (v367/v368, v376–v379), with the irreducible core $\{\pi, v_{\rm geo}\}$ a theorem (v160–v165). That residual is since sharpened to a single point: net existence and full-cone reflection positivity are discharged to **[E]** (v175), the realisation is assembled as one central theorem (v176) and split into the marks + kernel obligations whose finite cores are closed (v177/v178); the two then unify into one conformal-realisation premise (v179) which finally reduces — via uniformisation, Kerékjártó and the order-4 Möbius classification — to the milder, non-circular seam-*isometry* premise `QGEO.ISO.01` (the carrier clock is an order-4 orientation-preserving isometry of the RP seam metric, v180), and finally — by equivariant uniformisation — to the strictly weaker conformal-*symmetry* / deck premise `QGEO.SYM.01` (v181). Below that the residual is *definitional*: the carrier μ_4 clock *is* the conformal deck of the seam (the seam's conformal structure is the μ_4-deck structure of the carrier). It is since driven to its sharpest, *non-circular* form (v194–v198): the raw seam DtN is RP-canonical (Osterwalder–Schrader on the quasi-free state), and the bedrock is cracked to the *state-invariance* $\omega \circ \rho = \omega$ — the DtN principal symbol commutes *exactly* with the clock, and Tomita–Takesaki gives $[\rho, \Lambda_\Sigma] = 0$ with no conformal-covariance assumption, removing the Bisognano–Wichmann circularity — the maximally-operational form of the same postulate. The bounded residual is then reduced once more (v201): writing the DtN as $\Lambda = |k| + M_f$ with M_f multiplication by the curvature $f(\theta)$, block-diagonality holds iff f is $\mathbb{Z}_4$-invariant, and a curvature sourced by the four μ_4 marks $f = \sum_j g(\theta - 2\pi j/4)$ *is* automatically $\mathbb{Z}_4$-invariant — so the marks being a μ_4 orbit (forced by v195) *force* the sub-principal symbol block-diagonal. The whole residual thus collapses to the *mark-locality* of the DtN (the seam flat away from the marks = the conformal-deck structure): a single definitional identification — not a finite computation, not reducible further without relabeling — left honestly **[O]**.

The deductive chain below the postulate is now machine-checked. The geometric normal form (Uniform: $z \mapsto iz$, $\sigma\rho\sigma = \rho^{-1}$, orbit$\to \mu_4$, multiplier order-4 $\Leftrightarrow \zeta = \pm\, i$; `FORM.QGEO.02`) and the conditional implication *mark-local*$\Rightarrow \omega \circ \rho = \omega$ (`FORM.QGEO.01`) are **Lean-formalised** (audit: pass, only the three standard kernel axioms; mirrors v177 and v201/v210). So everything *downstream* of the postulate is **[E]**; the conformal-deck premise `QGEO.SYM.01` is itself the downstream face of the single keystone `SEAM.EQUIV.01` (next paragraph), which carries the one irreducible **[O]**.

Bisognano–Wichmann: the deck postulate is downstream of the seam being a chiral net ([`verification/v323_bw_geometric_modular.py`]). One further step links the two open bedrock items. The μ_4 clock is *literally* a geometric rotation: $\rho = \mathrm{diag}(i^n) = \exp(i\frac{\pi}{2}L)$ with $L = \mathrm{diag}(n)$ the seam rotation generator, so $\mu_4 \subset U(1)_{\rm rot}$ is the quarter-turn subgroup. **[E]** For a rotation-covariant seam covariance $C = f(L)$ the modular Hamiltonian $K = \log((1-C)/C) = g(L)$ is itself a function of L, so the modular flow $\sigma_t = e^{iKt}$ commutes with *all* rotations and the μ_4 clock is a modular symmetry *for free* — with no fine-tuned curvature (the discriminator: a mere period-4 curvature still preserves the clock, but its flow is *not* geometric, $[K, L] \neq 0$). **[C]** This is the Bisognano–Wichmann content: *given* the seam is the $(E_8)_1$ chiral conformal net (v308/`SEAM.EQUIV.01`), BW/Hislop–Longo make the vacuum modular flow geometric, so $\omega \circ \rho = \omega$ *follows* — `QGEO.SYM.01` is not an independent axiom but downstream of `SEAM.EQUIV.01` plus a rotation-invariant vacuum. **[O]** The single remaining premise is then that the raw collar vacuum is rotation-invariant (no preferred seam angle) — cleaner than a bespoke thermal state, and *shared* with the keystone, so the two open items collapse toward one (and the keystone itself is now *closed modulo a cited theorem* via the lattice/S_3 pinning and `FORM.SEAM.MMST.01`, only the cited continuum existence staying **[O]**). On the exactly solvable four-interval realisation of the four marks the invariance is *manifest* at the state level ($\rho C \rho^{-1}$=C at machine precision, with the fermionic clock the order-8 double-cover lift ρ^4=-1; v480) — the mechanism in the multilocal free-fermion class, the premise on the raw collar unchanged. This collapse is machine-pinned in Lean (`FORM.QGEO.BW.01`): `bedrockReducesToRotationInvariance` composes the Bisognano–Wichmann steps over the v308 chain, and `#print axioms` confirms the residual is exactly the cited theorems + the carrier recovery gap + the one shared premise (rotation-invariant vacuum).

The seam keystone theorem — closed modulo a cited theorem [E]/[C] ([`verification/v325_pillowcase_keystone.py`])

The whole bedrock can now be stated as *one* citable theorem with *exactly one* open hypothesis. **Hypothesis** [O] (H): the raw RP-collar seam state is the flat $\tau{=}i$ pillowcase state — equivalently, the OS-reconstructed seam vacuum is the rotation-invariant quasi-free state. **Then** [E]: the four marks are square ($n{=}2\chi(S^2){=}4$, v216; cross-ratio $2 \Rightarrow j{=}1728$, the order-4 CM modulus, v214/v267); the orbifold $S^2(2,2,2,2)$ has $\chi_{\text{orb}}{=}0$, i.e. it is flat at $\tau{=}i$ (Troyanov); the DtN sub-principal is supported only at $m \equiv 0 \bmod 4$, so $[\rho,\Lambda]{=}0 \Rightarrow \omega\circ\rho{=}\omega$ (`QGEO.SYM.01`, v201/v280/v323); and a holomorphic $c{=}8$ chiral net is $(E_8)_1$ ($\det\text{Cartan}(E_8){=}1$ vs $D_8{=}4$, v83/v143/v308, `SEAM.EQUIV.01`), with the carrier recovery gap $\Delta{=}6\ln\frac{3}{2}>0$ (v76/v302) as the input. By v323 the single hypothesis H is *shared* by both bedrock IDs. H is now pinned at every computable level by the explicit lattice model (v367/v368) and the S_3 stack ($c{=}8$, 248 currents/1 primary, torus GSD=1, RP; v376–v379) and is machine-pinned in Lean (`FORM.SEAMEQUIV.01` + `FORM.SEAM.MMST.01` + `FORM.QGEO.BW.01`), so the keystone is *closed modulo a cited theorem*: the *only* residual that stays [O] is the abstract continuum existence of the scaling limit (the cited MMST/OS theorems, v336), not H as a from-scratch construction.

The OS step, discharged at the many-body level ([`verification/v329_os_gap_reduction.py`]). The one analytic input the keystone chain still cited — "the OS-reconstructed bulk gap = the transfer gap" — is now discharged on the finite model. [E] The OS/Euclidean many-body Hamiltonian is the second quantization $d\Gamma(-\log T)$ of the recovery transfer T, whose spectrum is the set of sums of single-mode energies; hence the many-body (bulk) gap equals the smallest positive single-mode energy $= 6\ln\frac{3}{2}$, *independent of the number of modes* — exactly "bulk gap = transfer gap". [E] Under H the rest follows ($C{=}f(L) \Rightarrow [\rho,K]{=}0 \Rightarrow \omega\circ\rho{=}\omega$; $\det K{=}1 \Rightarrow$ holomorphic $\Rightarrow (E_8)_1$), so H is the *single sufficient premise* of both bedrock IDs. (The carrier piece of the holomorphy discriminator is itself now Lean-grounded in real determinants, `FORM.CARTAN.DET.01`: $|\det\text{Cartan}(A_3)|{=}|\det\text{Cartan}(D_5)|{=}4$, carrier index 16; and the rank-8 holomorphy discriminator $|\det\text{Cartan}(E_8)|{=}1$ — E_8 has one anyon — is now proved principally via the integer inverse, where Mathlib itself leaves $\det E_8{=}1$ a `proof_wanted`.) [O] The residual is only the *necessity* of H on the raw collar.

Necessity = the order-4 CM selection ([`verification/v331_necessity_of_H.py`]). The necessity of H is not vague: a concrete Steklov (Dirichlet-to-Neumann) computation shows the seam DtN $\Lambda{=}|k|{+}M_f$ commutes with the μ_4 clock ρ *if and only if* the four marks sit at the square $\{0,\frac{\pi}{2},\pi,\frac{3\pi}{2}\}$ (moving one mark off the square breaks the $\mathbb{Z}_4$ Fourier symmetry and $[\rho,\Lambda]\neq 0$). [E] And the square is distinguished: its cross-ratio is 2, i.e. $j{=}1728$, the unique 4-point configuration with an order-4 automorphism (v214/v267). So $H \Leftrightarrow$ square $\Leftrightarrow$ order-4 CM $\Leftrightarrow$ the μ_4 deck, and "necessity of H" sharpens to the single statement *the raw dynamics places the four Gauss-Bonnet marks (v216) at the order-4 CM point.* [O] Why the dynamics selects that point (vs a generic 4-puncture torus) is the irreducible residual. (The mark-locality criterion behind this — offset-4 couplings commute with ρ, offset-2 do not — is itself now exact Lean linear algebra over $\mathbb{Z}[i]$, `FORM.MU4COMM.01`.) The order-4 uniqueness (S fixes $\tau{=}i$, the unique order-4 elliptic point) and the fact that the square is the *Fekete attractor* of symmetric mark-equilibration are machine-checked in [`verification/v333_cm_selection.py`], sharpening the residual to "the deck acts geometrically".

The transfer functor as a runnable solver suite ([`verification/v326_ftransfer_suite.py`]). The four F_{transfer} readouts are productised into one harness, each a gapped relaxation to a unique attractor: [E] F_{pole} (Koide) is a Möbius contraction to $Q^*{=}2/3$ with multiplier exactly $(2/3)^6$ (the seam-clock subleading eigenvalue) — the only one at the *seam* rate; [C] $F_{\text{Boltzmann}}$ (η_B) is a washout contraction to the balance (H-theorem, thermal rate); [C] F_{relic} (axion) freezes the comoving number $N{=}E/\omega$ (the adiabatic invariant, cosmological rate); [E] F_{QCD} (m_p/m_e) is the 1-loop RG to the Gaussian UV fixed point with the carrier $b_0{=}7$ (Λ_{QCD} dynamically generated); and the F_{RareKaon} bridge lands on NA62 (v202). One shape, one seam rate, four external rates honestly fenced (v187/ v303) — the suite makes the frontier measurable without claiming the external rates as compiler outputs.

And the whole boundary QFT collapses onto that one postulate (the Mod-

ular Spectral Closure, [verification/v261_modular_spectral_closure.py]**).** The emergent-QFT round assembles the boundary theory into a single relative object $\mathsf{TFPT}_{\mathrm{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\mathrm{rel}})$: the finite Dirac is a covariance induction of the seam KMS state ([verification/v258_dirac_covariance_induction.py]), the spectral-action cutoff *is* that KMS weight ($f_2/f_0{=}1$, [verification/v259_modular_cutoff_kappa.py]), and the seam, the carrier-16 and E_8 are facets of one Kummer/K3 surface ([verification/v260_k3_kummer_unification.py]). A cross-consistency certificate pins the round ($4 = [B{:}A] = |\mu_4| = 2\chi = |(\mathbb{Z}/2)^2|$; one carrier-16; one gap $6\log\frac{3}{2}$, [verification/v261_modular_spectral_closure.py]). So Dirac, cutoff, gauging, glue and orientability are not separate residuals — they are readouts of the *same* seam state, and the boundary QFT is one relative object closed modulo the keystone `SEAM.EQUIV.01`, adding *no new structural class* (the inventory's "no sixth class" is preserved). *What the closure does* not *reach* stays exactly the dimensionful/ambient remainder: the full measured pole masses ($\times\, v_{\mathrm{geo}}$ + standard RG), the absolute QCD scales (m_p/m_e), the complete PMNS dynamics, and the *real* $4D$ *interacting* ambient measure $\mathrm{QG}_{\mathrm{amb}}$ — the F_{transfer} and ambient items already on the list.

The ambient measure on the decoupled sector ([verification/v330_qgamb_admissible_measure.py]). On the *admissible* sector the ambient measure is not just roadmapped (v275) but *constructed*: the seam transfer is reflection-positive (symmetric PSD), the gap gives exponential clustering ($C(n)/C(n{-}1) \to (2/3)^6$), and the finite susceptibility $\chi = \sum_n C(n) = 729/665$ makes $\mathrm{Var}(S_\Lambda)/\Lambda \to \chi$ converge, so the finite-volume measures are uniformly *tight* (Prokhorov) and the projective limit $\mu = \lim_\Lambda \mu_\Lambda$ EXISTS; OS reconstruction then gives $H_{\mathrm{OS}}{=} -\log T \geq 0$ with gap $6\ln\frac{3}{2}$ [E]. The construction *is* the gap (an ungapped transfer sends $\chi \to \infty$ and tightness fails). [C] The *non-admissible* (metric) sector — the full $\mathrm{QG}_{\mathrm{amb}}$ measure — is now discharged as a *redundancy* (v369, conditional on `SEAM.EQUIV.01`+Bisognano–Wichmann), a certification object rather than missing dynamics; by the gap-decoupling margin $\Delta - 31/(4\pi^2) \approx 1.648 > 0$ (v76) the physical readouts do not need it.

The metric-sector obstruction is the conformal mode ([verification/v332_qgamb_metric_sector.py]). The open "metric sector" is not a vague gap: it is precisely the Gibbons–Hawking–Perry conformal-factor problem. [E] The conformal mode $h_{\mu\nu}{=}\phi\,\eta_{\mu\nu}$ has a *negative* kinetic coefficient $c_{\mathrm{conf}}(D){=}-(D{-}1)(D{-}2)/4$ ($=-\frac{3}{2}$ at $D{=}4$), so the bare Euclidean weight $e^{-S} \sim e^{+|c|\phi^2}$ diverges over that direction — the measure obstruction. [E] But it lives in the conformal/gauge direction, *not* the admissible sector (positive and gapped, v330), so by the same gap-decoupling margin every physical readout is insulated from it. [C] *Perturbative* graviton unitarity is moreover established: the Stelle ghost is a Seeley–DeWitt *truncation artefact* — the entire form factor $e^{\Box/M^2}$ of v304 dresses the conformal propagator with no new pole, the spin-2 sector is ghost-free via the entire KMS form factor and the Barnes–Rivers decomposition (v370), and the truncation tower decouples (the nearest truncation-zero modulus runs to infinity, v380); only the polynomial $R{+}R^2$ truncation ever re-introduces the ghost, as a control. [C] The remaining *nonperturbative* ambient measure is the [C] redundancy above (v369); the conformal mode's nonperturbative resolution (a GHP contour rotation or the IDG resummation) is the conditional ingredient — the GHP sign-flip (the conformal Gaussian made convergent) and the IDG pole-free dressing together give a *conditional* mode-by-mode construction, machine-checked in [verification/v334_conformal_resolution.py].

Appendix: computational verification

Every [E] statement here is machine-checked in the TFPT suite and re-derived independently in Wolfram:

script	content
`v53_compiler_core.py`	$(5,3)$ skeleton, Pythagorean Δ_Y split, anchor char-poly uniqueness
`v54_seam_horizon_keystones.py`	triply-forced 8; shared transport $\lambda_2{=}(2/3)^6$; de Sitter constants
`v55_coxeter_cycle.py`	computed E_8 Coxeter element (order 30), exponents = totatives, $S_{dS}\rho_\Lambda{=}32\pi^4$
`v56_unique_attractor.py`	gapped transport ⇒ unique attractor, rate $(2/3)^6$; Coxeter planes; rank-1 reset

Run `python verification/run_all.py` (all pass) and, for the independent second path, `wolframscript -file verification/wolfram/tfpt_readouts.wl` (all pass). Figures are produced by `verification/make_figures.py`.

External works built upon

The TFPT discrete core (the two axioms, the E_8 glue, the integer skeleton) is self-contained and machine-checked. Its *bridges* to continuum/AQFT physics, to gravity and to the lattice/singularity picture build on established mathematics, listed here by topic; the body refers to these by author name.

Modular theory and algebraic quantum field theory.

- M. Takesaki, *Tomita's Theory of Modular Hilbert Algebras and its Applications*, Lecture Notes in Math. **128**, Springer (1970).
- J. J. Bisognano, E. H. Wichmann, *On the duality condition for a Hermitian scalar field / for quantum fields*, J. Math. Phys. **16** (1975) 985–1007; **17** (1976) 303–321.
- K. Osterwalder, R. Schrader, *Axioms for Euclidean Green's functions* I, II, Comm. Math. Phys. **31** (1973) 83–112; **42** (1975) 281–305.
- S. Doplicher, R. Haag, J. E. Roberts, *Local observables and particle statistics* I, II, Comm. Math. Phys. **23** (1971) 199–230; **35** (1974) 49–85.
- R. Haag, *Quantum field theories with composite particles and asymptotic conditions*, Phys. Rev. **112** (1958) 669–673; D. Ruelle, *On the asymptotic condition in quantum field theory*, Helv. Phys. Acta **35** (1962) 147–163.
- R. Longo, *Index of subfactors and statistics of quantum fields* I, II, Comm. Math. Phys. **126** (1989) 217–247; **130** (1990) 285–309.
- Y. Kawahigashi, R. Longo, M. Müger, *Multi-interval subfactors and modularity of representations in conformal field theory*, Comm. Math. Phys. **219** (2001) 631–669.

Conformal nets, vertex operator algebras and lattice constructions.

- I. B. Frenkel, V. G. Kac, *Basic representations of affine Lie algebras and dual resonance models*, Invent. Math. **62** (1980) 23–66; G. Segal, *Unitary representations of some infinite-dimensional groups*, Comm. Math. Phys. **80** (1981) 301–342.
- T. J. Osborne, A. Stottmeister, *Conformal field theory from lattice fermions*, Comm. Math. Phys. **398** (2023) 219–289, `arXiv:2107.13834`, `doi:10.1007/s00220-022-04521-8`. (This is the seam scaling-limit theorem the TFPT continuum leg cites — the Koo–Saleur → Virasoro convergence, $c = \frac{1}{2}$, and the smeared WZW-current Theorem 5.1. The internal TFPT code-token "MMST" and the legacy claim-ids `SEAM.MMST.*` / filenames `v*_mmst_*` denote *this* Osborne–Stottmeister theorem.)
- V. Morinelli, G. Morsella, A. Stottmeister, Y. Tanimoto, *Scaling limits of lattice quantum fields by wavelets*, Comm. Math. Phys. **387** (2021) 299–360, `arXiv:2010.11121`, `doi:10.1007/s00220-021-04152-5`; and *Operator-algebraic renormalization and wavelets*, Phys. Rev. Lett. **127** (2021) 230601. (The operator-algebraic (wavelet/Wilson–Kadanoff) renormalization *framework* the Osborne–Stottmeister theorem above builds on.)
- M. S. Adamo, Y. Moriwaki, Y. Tanimoto, *Osterwalder–Schrader axioms for unitary full vertex operator algebras*, `arXiv:2407.18222` (2024).

- M. S. Adamo, L. Giorgetti, Y. Tanimoto, *Rational and non-rational two-dimensional conformal field theories arising from lattices*, arXiv:2506.01008 (2025); *Representations of conformal nets associated with infinite-dimensional groups*, arXiv:2508.07109 (2025).
- P. Naaijkens, D. Penneys, B. Wallick, *(commuting-projector reflection positivity / LTO-RP)*, arXiv:2605.10693.

Lattices and quadratic forms.

- J. H. Conway, N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, 3rd ed., Grundlehren der math. Wiss. **290**, Springer (1999).
- C. L. Siegel, *Über die analytische Theorie der quadratischen Formen* (the Minkowski–Siegel mass formula), Ann. of Math. **36** (1935) 527–606.

Singularities and geometry.

- E. Brieskorn, *Beispiele zur Differentialtopologie von Singularitäten*, Invent. Math. **2** (1966) 1–14.
- J. Milnor, *Singular Points of Complex Hypersurfaces*, Ann. of Math. Studies **61**, Princeton Univ. Press (1968).

Spectral geometry, gravity and thermal time.

- A. H. Chamseddine, A. Connes, *The spectral action principle*, Comm. Math. Phys. **186** (1997) 731–750.
- A. Connes, C. Rovelli, *Von Neumann algebra automorphisms and time–thermodynamics relation in general covariant quantum theories*, Class. Quantum Grav. **11** (1994) 2899–2918.
- R. M. Wald, *Black hole entropy is the Noether charge*, Phys. Rev. D **48** (1993) R3427–R3431.
- H. Casini, M. Huerta, R. C. Myers, *Towards a derivation of holographic entanglement entropy*, JHEP **05** (2011) 036, arXiv:1102.0440.